Journal of Energy Storage 68 (2023) 107713

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Review Article

A review on thermal energy storage with eutectic phase change materials: Fundamentals and applications

Mingyang Sun [a], Tong Liu [a], Haonan Sha [a], Mulin Li [a], Tianze Liu [a], Xinlei Wang [a], Guijun Chen [a], Jiadian Wang [b,*], Dongyue Jiang [a,*]

[a] Key Laboratory of Ocean Energy Utilization and Energy Conservation of Ministry of Education, Dalian University of Technology, 116024, China

[b] No. 703 Research Institute of CSSC, 150000, China

ARTICLE INFO

Keywords:
Thermal energy storage
Phase change materials
Eutectic
Form-stable
Thermal conductivity

ABSTRACT

The storage and use of thermal energy have gained increasing attention from various countries. Phase change materials (PCMs) are commonly used in thermal energy storage (TES) applications due to their high latent heat. More than a hundred single-component PCMs have been reported, each with a specific phase change temperature. In addition to single-component PCMs, eutectic phase change materials (EPCMs) are also used in TES. EPCMs are homogeneous mixtures of binary, ternary, or multiple-component PCMs that melt at a unique temperature lower than the melting point of any of the constituents. Compared to single-component PCMs, EPCMs demonstrate superior performance in TES applications, with a broader phase change temperature range and improved phase stability. This review provides an overview of over 250 organic/inorganic EPCMs from eutectic theory, material classification, thermophysical properties, preparation and characterization methods, techniques for thermal performance improvement, and applications. The review also concludes with a discussion of future prospects and findings that EPCMs can enhance heat storage density and thermal stability when compared to single-component PCMs. This information serves as a reference for the development of heat storage materials with improved thermal physical properties in the future. Overall, this review provides a comprehensive summary of EPCMs from their fundamentals to applications.

1. Introduction

A TES system is essential for balancing energy supply and demand, even when they are mismatched in time and space. This system facilitates the storage of thermal energy from sources such as solar, geothermal, and industrial waste heat, to be used in various applications including power generation, water heating, building thermal comfort, battery thermal management (BTM), and more. Among the various thermal storage techniques, the latent heat thermal storage technique is preferred over sensible and chemical reaction-based storage systems because of its large heat of fusion and constant temperature during heat absorption and release. Hundreds of types of single-component organic and inorganic PCMs have been reported with specific phase change temperatures, as previously reviewed [1]. However, single-component PCMs face three major challenges in practical applications. Firstly, the phase change temperature of the single-component PCM may not be suitable for specific applications. For instance, a biocompatible PCM with a phase change temperature close to the physiological temperature of the human body (37 °C) is required for near-infrared triggered drug release. However, the phase change temperature of common biocompatible fatty acids may not meet this requirement. Secondly, single-component salt hydrates often face the issue of phase segregation. For example, $Na_2HPO_4 \bullet 12H_2O$ melts incongruently at the phase change temperature, forming $Na_2HPO_4 \bullet 7H_2O$ solid particles and Na_2HPO_4 aqueous solution. $Na_2HPO_4 \bullet 7H_2O$ sinks to the bottom of the container due to a greater density, and upon heat release, $Na_2HPO_4 \bullet 12H_2O$ forms in the solution and sinks to the bottom of the container. The $Na_2HPO_4 \bullet 7H_2O$ solid particles at the bottom are covered by a layer of $Na_2HPO_4 \bullet 12H_2O$, which requires a very long time for $Na_2HPO_4 \bullet 12H_2O$ to form through molecular diffusion. With repeated thermocycling, more $Na_2HPO_4 \bullet 7H_2O$ solid particles accumulate, and the thermal storage capacity decreases to one eighth (from 200 kJ/kg to 25 kJ/kg) after only four freeze-thaw cycles [2]. Thirdly, thermal decomposition is another limitation of single-component PCMs. For example, the thermal decomposition of dicarboxylic acids (adipic acid, suberic acid, sebacic acid, etc.) occurs at a temperature very close to their melting temperature, causing unexpected weight loss of the PCM in the thermal storage

* Corresponding authors.
E-mail addresses: wangjiadian@126.com (J. Wang), jiangdy@dlut.edu.cn (D. Jiang).

https://doi.org/10.1016/j.est.2023.107713
Received 16 February 2023; Received in revised form 9 April 2023; Accepted 13 May 2023
Available online 20 May 2023

Nomenclature

Phase Change Material	PCM
Eutectic Phase Change Material	EPCM
Thermal Energy Storage	TES
Battery Thermal Management	BTM
Concentrated Solar Power	CSP
Ammonium Polyphosphate	APP
Char Forming Agent	CFA
Polymethyl Metharcrylate	PMMA
n-heptadecane	C17
n-octadecane	C18
n-nonadecane	C19
n-eicosane	C20
n-tetracosane	C24
n-octacosane	C28
Capric Acid	CA
Decanoic Acid	DA
Dodecanoic Acid	DDA
Hexadecanoic Acid	HA
Lauric Acid	LA
Myristic Acid	MA
Oleic Acid	OA
Palmitic Acid	PA
Stearic Acid	SA
Differential Scanning Calorimetry	DSC
1-dodecanol	DE
Tetradecanol	TD
Polyethylene Glycol	PEG
Expanded Graphite	EG
Carbon Nanotubes	CNTs
Multi-Walled Carbon Nanotubes	MWCNTs
Carbonized Abandoned Rice	CAR
Photovoltaic	PV
Trimethylethane	TME
Sodium Thiosulfate Pentahydrate	STP
Sodium Stearate	SS
Silica	SiO_2
Near-infrared Radiation	NIR
Disodium Hydrogen Phosphate Dodecahydrate	DHPD
Sodium Acetate Trihydrate	SAT
Direct Steam Generation	DSG

system. These challenges are commonly faced by single-component PCMs, and they limit their applicability and long-term operation in practical applications. A promising method to solve these challenges is highly desired. Some challenges (e.g., phase segregation) can be addressed by methods such as the use of thickening agents, water addition, and microencapsulation (as detailed in previous review articles [3]), but these methods significantly reduce the capacity of thermal energy storage.

EPCMs are homogeneous mixtures of binary, ternary or *n*-nary PCMs that can be completely blended in the liquid state [4]. They melt at a single temperature that is lower than any of the constituents. This property makes EPCMs ideal for targeting phase change temperature ranges that single-component PCMs cannot reach. For instance, a eutectic mixture of lauric acid (melting point 44 °C) and stearic acid (melting point 69 °C) at a weight ratio of 4:1 can yield a phase change temperature of 39 °C. This temperature is very close to the physiological temperature of the human body, making it suitable for near-infrared triggered drug release applications [5]. EPCMs made with binary Na_2CO_3•$10H_2O$-Na_2HPO_4•$12H_2O$ salt hydrates modulate the solubility of the constituents. This leads to the formation of a form-stable EPCM at a ratio of 40 wt% Na_2CO_3•$10H_2O$-60 wt% Na_2HPO_4•$12H_2O$. The EPCM has a high phase change enthalpy of 220 kJ/kg [6] and maintains thermal stability for up to 200 cycles [7]. Another way to address the limitations of single-component PCMs is by adding stearic acid to single-component dicarboxylic acids (adipic acid, suberic acid, sebacic acid) to form EPCMs. This process reduces the phase change temperature by 60–80 K compared to the single-component PCMs. This large reduction in the phase change temperature prevents the thermal decomposition of the dicarboxylic acids. Although the EPCM have these advantages, the EPCM itself is obtained by blending a variety of single-component PCMs, and therefore it still faces problems such as low thermal conductivity, phase separation, supercooling and corrosion.

Currently, over 250 EPCMs have been reported for various heat storage applications. Previous reviews have summarized some of these EPCMs, such as fatty acids [8], high-temperature metal alloys, and molten salts [9,10]. However, these reviews did not include EPCMs reported in recent years. The article [1] briefly mentioned EPCM and focused on improving the energy storage characteristics of PCMs through encapsulation and nanomaterial additives without elaborating on the classification and theory of EPCM. Review [11] only summarized the application of EPCMs in energy storage in the construction field. Another review [12] summarized the thermal management of PCMs used in electronic devices and discussed methods to improve their thermal conductivity. Similarly, review [13] summarized the application of PCM in recent years without elaborating on the theory of eutectic and EPCMs. Furthermore, these review studies were not organized in terms of eutectic systems and did not explain the pros and cons of using EPCMs. A recent review mentioned eutectic materials [10], but this review only studied organic EPCMs, and only considered the effect of thermal conductivity on EPCMs, without studying inorganic EPCMs and the supercooling and phase separation during phase transition. This paper summarizes different types of EPCMs from the aspects of eutectic theory and phase diagram, elaborates on the advantages and disadvantages of using EPCMs, and summarizes their application in different situations. It provides directions for future researchers to prepare EPCMs with greater heat storage and higher thermal stability, summarizes the research progress of PCMs, and looks forward to future research in this area.

This study presents a comprehensive review of reported EPCMs, covering both low-to-moderate and moderate-to-high temperature ranges. The review includes binary and ternary EPCMs formed by organic PCMs, inorganic PCMs (including alloys, salt hydrates, and molten salts), as well as a combination of the two. Starting with the eutectic theory (Section 2), the review goes on to cover the thermophysical properties of the reported organic and inorganic EPCMs (Section 3). Section 4 focuses on methods for enhancing the performance of EPCMs, specifically in terms of thermal conductivity and thermal stability. Section 5 introduces diverse thermal energy storage applications using EPCMs. Finally, Section 6 raises perspectives for future study on EPCMs.

2. Eutectic system

Different from the single-component PCMs, the melting point of an EPCM is affected by the composition of each constituent. The relationship between the melting temperature and composition is important for EPCMs. Phase diagrams are typically employed for describing such a relationship. The phase diagram could be plotted by Schroeder equation which is derived from the change in free Gibbs enthalpy:

$$G = H - TS \Rightarrow \begin{cases} H = G + TS \\ \left(\frac{\partial G}{\partial T}\right)_P = -S \end{cases} \Rightarrow H = G - T\left(\frac{\partial G}{\partial T}\right)_P \quad (1)$$

In which G represents the Gibbs free enthalpy (J/mol), H represents enthalpy (J/mol), T represents temperature (K), S represents entropy (J/mol/K), and P refers to pressure. The derivative of G/T at constant pressure could be derived as:

$$\left(\frac{\partial G/T}{\partial T}\right)_P = -\frac{1}{T^2}\left(G - T\left(\frac{\partial G}{\partial T}\right)_P\right) = -\frac{H}{T^2} \quad (2)$$

By assuming the activity is equal to the concentration, the chemical potential μ_i can be expressed as:

$$\mu_i = \mu_i^0 + RTln\frac{a_i}{a} \approx \mu_i^0 + RTlnx_i \quad (3)$$

In which R is the ideal gas constant (8.314 J/mol/K), a is the activity (J/mol) and x is the molar fraction. At the equilibrium condition, μ_i equals to zero and μ_i^0 could be expressed as $\mu_i^0 = -RTlnx_i$. Integrating the relationship in Eq. (2), the μ_i/T derivative could be expressed as:

$$\left(\frac{\partial \mu_i/T}{\partial T}\right)_P = \frac{\partial}{\partial T}(Rlnx_i) \Rightarrow Rlnx_i = -\frac{H_i^0}{T} + K \quad (4)$$

In which H_i^0 represents the enthalpy of fusion (J/mol), and K is the integration constant. K could be determined for a pure component. By substituting $x_i = 1$ into Eq. (4), K could be calculated as H_i^0/T_i^0, T_i^0 is the phase change temperature. The relationship between molar fraction x_i and temperature T could be obtained as:

$$Rlnx_i = -\frac{H_i^0}{T} + \frac{H_i^0}{T_i^0} \quad (5)$$

For a binary EPCM with different molar ratios of A and B, a phase diagram could be plotted by substituting the ideal gas constant R, phase change temperature T_i^0 and heat of fusion H_i^0 of each component into Eq. (5). The plotted phase diagram is illustrated in Fig. 1a. The x axis shows the composition of the EPCM, where the left most point represents pure A and right most point represents pure B. The pressure of the system is assumed as constant and the temperature is plotted on the y axis. The red and blue profiles which separate liquid phase with "A + Liquid" and "B + Liquid" are liquidus curves, while the black profile separating the solid phase with "A + Liquid" and "B + Liquid" is solidus curve. The intersection points of the liquidus and solidus curves is the eutectic point. At the eutectic point, the solid A, solid B and liquid all exist in equilibrium.

The crystallization of a composition X which contains 80 % B and 20 % A is discussed as an example. When the temperature is above T_1, the composition X would be in liquid phase. When the temperature drops to T_1, crystal B starts to form in the liquid. With a further decrease in temperature, a larger amount of B crystalizes and the liquid composition contains a greater molar fraction of A. In Fig. 1(a) with the temperature reduction from T_1, T_2, T_3, T_E, the liquid composition will change from point 1, 2, 3, to the eutectic point. There are two phases during this temperature reduction including crystalized B and liquid. When the temperature is reduced to T_E, solid A starts to crystalize. When a complete crystallization is achieved, the mixture of 20 % solid A and 80 % solid B is formed. During the crystal's formation process, the molar fraction of solid B as well as liquid could be determined by the lever rule. When the temperature is reduced to T_2, the percentage of crystal B could be determined by b/(a + b) and the percentage of liquid could be determined by a/(a + b). When the temperature further drops to T_3, more crystal B formed and the percentage of B increases to d/(c + d).

For the melting process of a the phase transition process -component PCM, heat is required to raise its temperature to the melting point initially, followed by extra heat to complete melting. This amount of extra heat is referred as heat of fusion or latent heat. When the solid and liquid are in thermodynamic equilibrium, the change in the free Gibbs enthalpy ΔG is zero during the melting process. The melting temperature of single-component PCM could be determined by:

$$T_{m-single} = \frac{\Delta H_{single}}{\Delta S_{single}} \quad (6)$$

As for the melting process of an EPCM, the change in enthalpy would be similar to that of melting a single-component PCM. The difference mainly originates from the change in entropy. As shown in Fig. 1b, the melting of an EPCM into liquid with multiple compositions would incur a greater disorder as well as larger change in entropy $\Delta S_{eutectic}$. This leads to a reduced melting temperature of the EPCM as compared to the single-component PCM which contributes to a wider temperature range for TES.

3. Eutectic phase change materials

There are a plethora of solid-liquid PCMs available for thermal storage applications, which can be classified as organic or inorganic based on their composition. Each type of PCM has its own advantages and drawbacks. For instance, organic PCMs typically have a high latent heat and are free of phase segregation, but they can be expensive and have poor thermal conductivity and chemical stability. In comparison, inorganic PCMs are relatively inexpensive and have a high latent heat, but they suffer from phase segregation (salt hydrates), a large degree of supercooling, and poor thermal conductivity. Based on either organic or inorganic PCMs, an EPCM can be synthesized between inorganic-

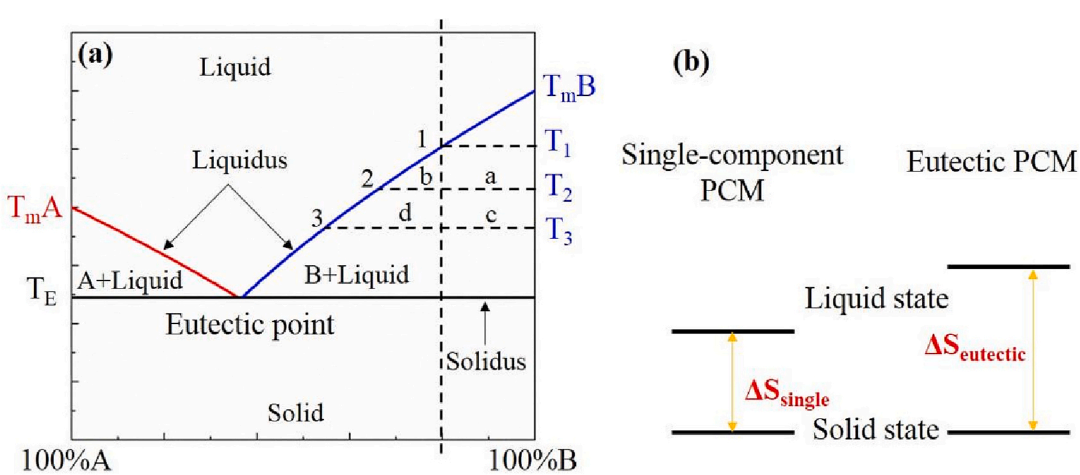

Fig. 1. (a) Phase diagram of EPCM and (b) Entropy change of single-component PCM and EPCM during melting process.

inorganic, organic-organic, or inorganic-organic, as illustrated in Fig. 2. This section will mainly focus on the reported binary or ternary EPCMs based on organic or inorganic PCMs.

3.1. Organic EPCMs

Organic PCMs, such as paraffin, fatty acids, and sugar alcohols, are commonly used natural materials for thermal storage applications, particularly in low-to-moderate temperature ranges. By combining different compositions of these PCMs, binary, ternary, or even *n*-nary EPCMs can be synthesized.

3.1.1. Paraffin based EPCMs

Paraffin is widely used in TES systems due to its lower or non-supercooling, lower vapor pressure, lower cost, and good recyclability. It has a temperature range between 46 and 68 °C and a large heat of fusion between 200 and 220 kJ/kg. Paraffin is non-reactive with common chemical reagents, making it chemically stable. Table 1 shows various paraffin-based EPCMs that have been synthesized and reported. Sarı et al., [14] prepared four paraffin-based EPCMs, including *n*-heptadecane (C17) - *n*-tetracosane (C24), *n*-nonadecane (C19) - *n*-octadecane (C18), *n*-nonadecane (C19) - *n*-tetracosane (C24), and *n*-eicosane (C20) - *n*-tetracosane (C24). This group of EPCMs shows a broadened melting point range between 19.24 and 35.8 °C. The eutectic ratios of the four EPCMs vary, as shown in Table 1. These EPCMs were synthesized by an emulsion polymerization method to form micro/nano polymethyl methacrylate (PMMA)/paraffin eutectic mixture encapsulations with a range from 1.16 to 6.42 μm. The synthesized paraffin eutectic and PMMA encapsulation exhibit improved thermal and chemical stability even after 5000 thermal cycles up to 160 °C. Li et al., [15] prepared an EPCM based on solid paraffin *n*-octacosane (C28) and liquid paraffin (mineral oil) and introduced ammonium polyphosphate (APP) and a char-forming agent (CFA) to form flame-retarded PCM. The flame-retarded PCM has a phase change temperature of 54 °C and a latent heat of 158 kJ/kg. With the incorporation of APP and CFA into the EPCM, good flame-retardant performance was achieved, reducing the peak heat release rate from 1189.7 to 135.9 kW/m^2, and suppressing the smoke production rate. Narayanan et al., [16] prepared a eutectic gel PCM formed by paraffin and oleic acid, which has a melting point of 54 °C and latent heat of 158 kJ/kg. By adding a minimal amount (0.5 wt%) of nanographite into the eutectic gel PCM, thermal conduction was significantly enhanced. The duration of the nanographite-enhanced eutectic gel PCM reduced to 3 mins, compared to the sample without nanographite addition, which requires 45 mins to melt.

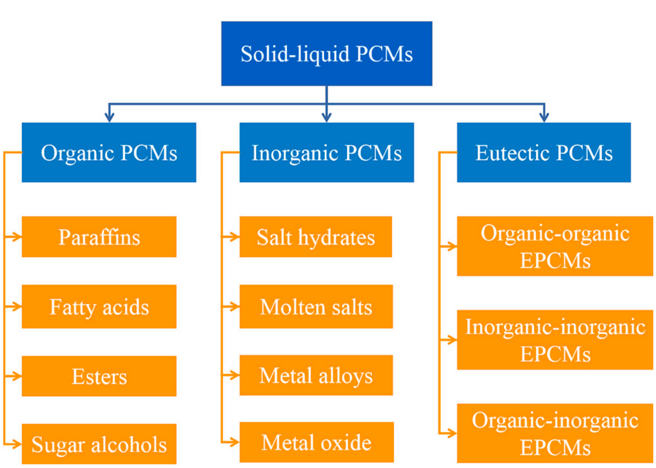

Fig. 2. Classification of solid-liquid phase change materials.

3.1.2. Fatty acids based EPCMs

Fatty acids are organic compounds made up of C, H, and O. They consist of a carbon chain with a methyl group at one end and a carboxyl group at the other [17]. Fatty acids are categorized as short-chain (<6 carbon atoms), medium-chain (between 6 and 12 carbon atoms), and long-chain (>12 carbon atoms) based on the length of the carbon chain. As illustrated in Fig. 3, fatty acids exhibit a broad range of phase transition temperatures, making them suitable for TES applications. As the carbon number increases, so does the phase change temperature and latent heat of the fatty acids. Fatty acids with carbon numbers ranging from 10 to 18 are commonly used as PCMs [8], including Caprylic acid, Capric acid (CA), Decanoic acid (DA), Dodecanoic acid (DDA), Hexadecanoic acid (HA), Lauric acid (LA), Myristic acid (MA), Oleic acid (OA), Palmitic acid (PA), and Stearic acid (SA). The thermal properties of these pure fatty acids for TES applications have been extensively studied and reported [8,18]. With over 10 potential fatty acids for use as PCMs, >120 binary, ternary, or n-nary EPCMs have been reported using combinations of these fatty acids or with other organic compounds.

Table 1 shows that Capric acid ($CH_3(CH_2)_8COOH$, CA) with a melting temperature of 32 °C is widely used to create EPCMs with LA, MA, PA, SA, Cetyl alcohol, and other compounds. Ke et al., [19] used the ultrasonification method to prepare a group of CA-based EPCMs and used Schroeder's equation to predict the eutectic point and mass fraction [20]. The predicted data matched well with differential scanning calorimetry (DSC) results. The lowest eutectic temperature of the EPCM will be significantly lower than the phase transition temperature of any single component, and at different component ratios, DSC will detect multiple peaks, with the first peak representing the lowest eutectic point transition temperature. Due to the relatively low melting temperature of CA, CA-based EPCMs have a temperature range of 13.94 to 27.8 °C and latent heat of 100.5 to 171.2 kJ/kg. In some studies, DA was used as a substitute for capric acid (EPCMs NO. 26-28). Lauric acid ($CH_3(CH_2)_{10}COOH$, LA) with a melting temperature of 44 °C is commonly prepared with MA, PA, SA, and other compounds to form EPCMs (EPCMs NO. 34-41). Hassan et al., [21] prepared a group of LA-based EPCMs by melt blending and ultrasonification. These EPCMs have a temperature range of 17 to 39 °C and latent heat between 90.2 and 205.4 kJ/kg. The largest latent heat of 205.4 kJ/kg was achieved when mixing LA with methyl palmitate at a weight ratio of 40:60. Myristic acid ($CH_3(CH_2)_{12}COOH$, MA) with a melting point of 52 °C is typically combined with fatty acids or other organic PCMs such as 1-dodecanol to form EPCMs (No. 43-49). A group of EPCMs with a majority of MA and a minority of SA or PA were reported in [22–24]. The melting point of the MA-based EPCMs varies from 18.43 to 46.7 °C, making them good candidates for low-temperature solar heating storage systems. The latent heat of the MA-based EPCMs varies from 128.6 to 205.3 kJ/kg, with the largest latent heat achieved when MA was mixed with Tetradodecanol at a weight ratio of 30.3:69.7. Stearic acid ($CH_3(CH_2)_{16}COOH$, SA) is widely used in TES applications due to its relatively higher melting point (~70 °C) and latent heat (180–210 kJ/kg). The SA-based EPCMs (No. 55-70) are typically formed between SA and fatty acids like PA, LA, MA, and other organic compounds like amides, acetanilide, and commercial PCM PT68. When mixed with fatty acids, the SA-based EPCMs have a relatively low melting temperature range from 27.4 to 32.2 °C. When the EPCMs are formed between dicarboxyl acids (Sebacic acid or Adipic acid) and SA, the melting point increases to 52.1–69.3 °C.

3.1.3. Other organic PCMs based EPCMs

Apart from fatty acids, sugar alcohols are also viable options for TES applications (No. 75-91). Sugar alcohols, also known as polyols or polyhydric alcohols, are low-digestible carbohydrates obtained by replacing aldehyde groups with hydroxyl groups. They are low-molecular-weight carbohydrates with an OH group on all carbon atoms, and can be described by a chemical formula: $C_nH_{2n+2}O_n$ [25]. Sugar alcohols, such as xylitol, sorbitol, erythritol, mannitol, inositol, and dulcitol, are attractive PCMs [26]. Among them, the single component sugar alcohol

Table 1
Organic eutectic phase change materials.

Material number	Phase change material	Composition (wt%)	Phase change temperature (°C)	Heat of fusion (kJ/kg)	Reference
Paraffin based EPCMs					
1	C17-C24	90:10	19.24	171.14	[14]
2	C19-C18	5:95	25.15	180.28	
3	C19-C24	95:5	32.2	187.9	
4	C20-C24	90:10	35.8	265.5	
5	C28-Mineral oil	1:1	21	129.8	[15]
6	Paraffin wax-OA	1:1	54	158	[16]
Fatty acids based EPCMs					
7	Caprylic acid:1-dodecanol(DE)	7:3	6.52	171.1	[30]
8	CA:LA:PA:SA	65.05:22.39:7.91:4.65	13.94	132.3	[19]
9	CA:LA:MA:PA:SA	61.1:24.6:8.1:4.0:2.2	14.04	132.8	
		50.7:31.2:10.3:5.1:2.7	17.64	132.8	[21]
10	CA:LA:MA:PA	63.1:19.7:12.4:4.8	14.61	133.4	[19]
11	CA:LA:MA:SA	62.1:23.1:11.4:3.4	15.13	129.6	
12	CA:LA:PA	63.4:31.5:5.1	15.47	134	
		61.9:31.0:7.1	21.7	100.9	[31]
13	CA:LA:MA	66.4:20.6:13.0	15.83	131.4	[19]
14	CA:LA:SA	30:60:10	16.4	174.9	[21]
		65.3:32.5:2.2	17.9	132.5	[19]
15	LA:DE	29:71	17	175.3	[32]
16	CA:MA:PA:SA	71.8:19.8:5.3:3.1	17.47	138.2	[19]
17	CA:MA:PA	71.8:18.9:9.3	17.56	136.8	
		64.8:22.6:12.6	17.7	148.7	[33]
18	CA:LA	64:36	18.5	162.9	[21]
		65:35	18.92	107.9	[34]
		65.1:34.9	19.1	126.5	[35]
		66.8:33.2	19.26	127.2	[19]
		67:33	22.81	154.2	[36]
19	CA:MA:SA	71.8:18.9:9.3	18.9	145.9	[19]
		64.8:22.6:12.6	23.9	158.7	[37]
20	CA: Tetradecanol (TD)	38:62	18.9	100.5	[38]
21	CA:PA:SA	79.3:14.7:6	18.5	139.3	[39]
		79.3:14.7:6	18.9	147.2	[40]
		83.8:10.2:6.0	19.7	145.7	[19]
		80:10:10	19.8	154.1	[21]
22	CA:MA	78.4:21.6	20.1	155.2	[19]
		78:22	20.5	153	[21]
		73:27	21.7	165.2	
		72:28	21.7	139.2	[41]
		72:28	18.2	148.5	[42]
		73:27	21.7	168.4	
		75:25	22.2	153.2	
		76:24	23.6	147.7	
		78:22	19.7	149	
		72:28	21.7	139.2	[43]
		73.6:26.4	22.6	154.8	[44]
		73.6:26.4	22.6	154.8	[45]
		72:28	25.4	139.2	[36]
23	CA:SA	86:14	24.1	161.4	[5]
		94.5:5.5	25.7	156.4	[19]
		77:23	27.8	122.6	[36]
24	CA:PA	76.5:23.5	21.85	171.2	[46]
		89:11	22.9	141.4	[19]
		88:12	29.6	163.7	[47]
25	CA:Cetyl alcohol	70:30	22.9	144.9	[48]
26	DA:DDA	75:25	19.9	163	[49]
27	DA:MA	78:20	20.5	153	[50]
		82:18	24.9	155	[49]
		71:29	33.9	182	[49]
28	DA:HA	85:15	28.1	156	[49]
29	DDA:OA	85:15	39.9	183	[49]
30	LA:DE	29:71	17	175.3	[32]
31	LA:Myristyl alcohol	40:60	21.3	151.5	[51]
32	LA:TD	46.4:53.6	24.5	90.2	[38]
33	LA: Methyl palmitate	40:60	25.6	205.4	[52]
34	LA:MA:SA	61:30.1:8.9	29.2	152.9	[19]
		57.5:34.3:8.2	29.4	140	[21]
35	LA:MA:PA	63.3:24.6:12.1	30.8	155.7	[19]
		55.3:29.7:15	31.4	145.8	[53]
36	LA:PA:SA	72.3:21:6.7	32	159	[19]
37	LA:SA	69:31	33.5	161.6	[54]
38	LA:PA	69:31	32.9	166.9	[55]
		77.5:22.5	33.6	169.6	[19]

(*continued on next page*)

Table 1 (*continued*)

Material number	Phase change material	Composition (wt%)	Phase change temperature (°C)	Heat of fusion (kJ/kg)	Reference
		69:31	35.2	166.3	[22]
		69:31	35.2	166.3	[21]
		77.5:22.5	35.7	171.8	[56]
39	LA:MA	63.6:36.4	33.3	173.6	[19]
		66:34	34.2	166.8	[22]
		66:34	34.2	166.8	[21]
		66:34	34.2	166.8	[57]
		67.7:32.3	34.6	163	[58]
		58:42	35.2	162.3	[36]
40	LA:1,6-hexanediol	70:30	36.9	177.1	[59]
41	LA:SA	86:14	37.5	199.6	[60]
		86.5:13.5	38.0	181.2	[19]
		75.5:24.5	37	182.7	[61]
		80:20	39	N.A.	[5]
42	LA:PT68	80:20	38.3	180.3	[21]
43	MA:DE	17:83	18.43	180.8	[32]
44	MA:TD	30.3:69.7	32.5	205.3	[62]
		28.2:71.8	33.2	128.6	[38]
45	MA:SA	64:36	44.1	182.4	[22]
		64:36	44.1	182.4	[21]
		64:36	44.7	N.A.	[23]
46	MA:PA	58:42	42.6	169.7	[63]
		65:35	45.4	183.1	[19]
		60:40	45.5	176.12	[64]
		70:30	46.7	155.4	[24]
		70:30	46.7	155.4	[65]
47	MA:PT68	69:31	46.3	182.2	[21]
48	MA:SA	76.3:23.7	46.4	180.6	[19]
49	MA:PA:SA	56.8:28.1:15.1	46.7	170.3	
50	OA: polyethylene glycol (PEG)	88:12	3.42	112.8	[66]
51	PA:DE	10:90	20.1	191	[32]
52	PA:1-hexadecanol	64.5:35.5	50.2	226.6	[67]
53	PA:PT68	58:42	54.9	184.5	[21]
54	PA:SA	64.2:35.8	52.3	181.7	[68]
		63:37	53.7	204.7	[19]
		62:38	53.9	177.7	[69]
		62:38	54	177.7	[70]
		58.8:41.2	54	177.7	[71]
		61:39	55.1	181	[21]
		63.2:36.8	55	180	
		62.9:37.1	56.2	204.7	
55	SA:PA:LA:MA	11.9:18.5:45.6:24	27.4	127.7	[21]
		6.8:12.6:55.2:25.4	29.6	151.9	
		5.6:10.3:63.2:20.9	29.6	151.9	[19]
56	SA:PA:LA:MA:PT68	17.4:11.1:42.9:22.6:6	27.8	160.8	[21]
57	SA:PA:LA:PT68	14:21.9:54:1	30	157.9	[21]
58	SA:PA:LA	6.8:21:72.2	32	159	[21]
		15.6:24.4:60	32.2	159.9	
59	SA:PA:PT68	43.9:28.1:28	52.1	167.9	[21]
60	SA:1, 10-decanediol	86:14	56.5	247.6	[72]
61	SA:hexanamide	38.6:61.4	58	176.6	[73]
62	SA:acetamide	76:24	62.23	222.1	[74]
63	SA:n-octanamide	82.7:17.3	63.1	199	[75]
64	SA:n-butyramide	70.9:29.1	64.3	198.4	
65	SA:benzamide	88:12	65.9	200.2	[76]
66	SA:suberic acid	96:4	66.4	191.4	[77]
67	SA:sebacic acid	96:4	67.1	196.6	
68	SA:adipic acid	98:2	67.5	200.3	
69	SA:PT68	47:53	68.7	210.6	[21]
70	SA:Acetanilide	90:10	69.3	220	[78]
71	Benzoic acid:Acetanilide	70:30	75.6	193.6	[79]
72	Boric acid:Oxalic acid dihydrate	12:88	87.3	344	[80]
73	Adipic acid:Sebacic acid	48:52	116	206	[81]
74	Adipic acid:Succinic acid	70:30	138.3	231	[82]
Other organic PCMs based EPCMs					
75	Xylitol:d-sorbitol	52:48	74.2	202.2	[28]
76	Xylitol:erythritol	75:25	83.9	261.2	
77	d-sorbitol:Erythritol	70:30	86.6	214.5	
78	Xylitol:D-mannitol	96:4	92.2	233.4	
79	Xylitol:Inositol	98:2	92.8	237.9	
80	Xylitol:d-dulcitol	99:1	93	238.5	
81	d-sorbitol:D-mannitol	95:5	97.5	169.9	
82	d-sorbitol:d-dulcitol	98:2	98.5	167.4	
83	d-sorbitol:Inositol	98:2	98.7	165.9	

(continued on next page)

Table 1 (*continued*)

Material number	Phase change material	Composition (wt%)	Phase change temperature (°C)	Heat of fusion (kJ/kg)	Reference
84	Erythritol:D-mannitol	84:16	113.1	324.1	
85	m-Erythritol:D-mannitol	87:13	114.4	322.8	[29]
86	Erythritol:d-dulcitol	95:5	117.1	332.4	[28]
87	Erythritol:Inositol	96:4	117.9	329.3	
88	Galactitol:Mannitol	30:70	153	292	[83]
89	D-mannitol:d-dulcitol	70:30	155	296.9	[28]
90	D-mannitol:Inositol	82:18	159.6	276.9	
91	d-dulcitol:Inositol	69:31	178.3	310.6	

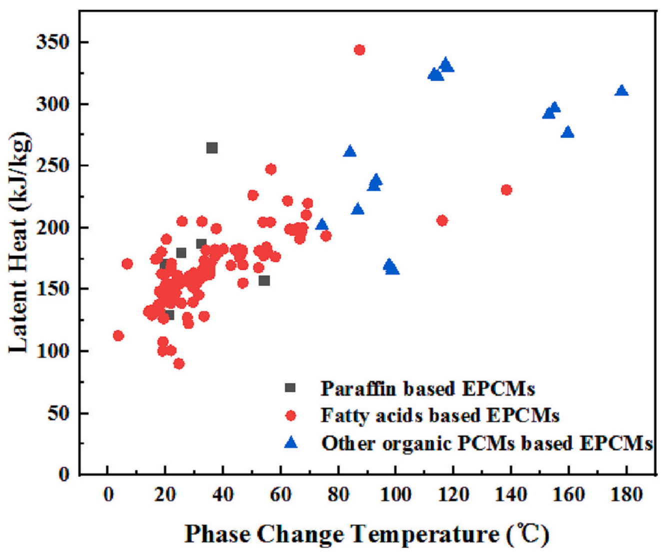

Fig. 3. Summary of organic EPCMs.

is suitable for the temperature range of 90 °C ~ 190 °C, it has a higher latent heat value of 180–350 kJ/kg.

Xylitol, with a phase change temperature of 93.3 °C and a latent heat value of 237.5 kJ/kg, has been widely noted for its large latent heat value, without phase separation, good economy and high cyclic stability [27]. Xylitol-based eutectic heat storage with other sugar alcohol PCMs (No. 75-80) has been found to have a reduced eutectic melting point and a small difference in latent heat value, but with low heat recovery efficiency due to the low nucleation rate and large supercooling of xylitol [28]. The results also showed that for xylitol and d-sorbitol based EPCMs, crystallization was difficult to occur even at a very low cooling rate of 0.5 °C. In addition to xylitol, the article [29] mentions the high latent heat value of erythritol (No. 84-87) and d-dulcitol (latent heat value >330 kJ/kg), but both also have a large supercooling problem. If the supercooling problem can be resolved, erythritol-based EPCMs with high latent heat value (about 330 kJ/kg) can be widely used. However, other sugar alcohol EPCMs (No. 88-91) with a large degree of supercooling, high eutectic temperature (above 150 °C), small latent heat value (<300 kJ/kg), poor cycle stability, and other issues have not yet achieved large-scale applications [28].

Fig. 3 summarizes the classification of existing EPCMs. The main phase transition temperature of EPCMs falls within the range of 20 °C to 80 °C, with latent heat values ranging from 120 kJ/kg to 250 kJ/kg. Among the organic EPCMs, fatty acid EPCMs occupy the highest proportion. For other sugar alcohol organic EPCMs, the latent heat value is larger, with most of them exceeding 200 kJ/kg. In comparison to paraffin-based EPCMs, the phase change temperature of these EPCMs is higher than 80 °C, while the phase change temperature is lower, typically ranging between 20 °C to 40 °C.

3.2. Inorganic EPCMs

Inorganic PCMs, which include metals, salt hydrates, and molten salts, have pervasive ionic bonding. Unlike organic PCMs, which are primarily used in the low-to-moderate temperature range, inorganic PCMs work across a wider temperature range. Inorganic PCMs typically have larger thermal conductivity and volumetric energy storage density than organic PCMs, making them important supplements, especially for high-temperature applications. Similar to organic PCMs, various EPCMs can form between metals, molten salts, and salt hydrates. These EPCMs exhibit different thermophysical properties compared to single-component inorganic PCMs.

3.2.1. Metal alloys

Metal and metal alloys are ideal for high-temperature TES applications due to their high melting temperature. Unlike organic PCMs, salt hydrates, and molten salts, metal alloys have thermal conductivities up to two or three orders of magnitude higher than other PCMs. This allows for rapid heat absorption and release in large power TES systems. However, as shown in Fig. 4, metal alloys have poor energy storage density per unit mass [84] (100–500 kJ/kg) compared to molten salt EPCMs (>600 kJ/kg), making them less attractive as TES candidates. Nevertheless, their high density and energy storage density per unit volume make them a viable alternative when space is a consideration. In addition to their superior thermal conductivity, low vapor pressure, and reduced degree of supercooling, metal alloys can also be synthesized to have a specific melting temperature for a particular application.

Pure metals and semiconducting materials like magnesium, aluminum, zinc, copper, and silicon have high melting temperatures compared to organic PCMs and salt hydrates. For example, the melting temperature of the lowest-melting metal, zinc, is as high as 419.5 °C [85], while the highest-melting material, silicon, has a melting point of up to 1410 °C [86]. Such high melting temperatures limit the widespread use of pure metals or semiconducting materials in TES. However, Risueño et al., [87,88] synthesized magnesium, zinc, and aluminum-based metal alloys for use in concentrated solar power (CSP) plants. Their results indicate that the metal alloy $Mg_{70}Zn_{24.9}Al_{5.1}$ and $Zn_{85.8}Al_{8.2}Mg_6$, with melting temperatures of 338 °C and 345 °C, respectively, are good candidates for CSP applications. These melting temperatures are much lower than those of pure metals, and the alloys maintain a high thermal conductivity of up to 59 W/(m·K) at high operating temperatures. In addition, these EPCMs did not show degradation after 700 thermal cycles, and energy-dispersive X-ray spectrometry tests showed no corrosion with four common stainless steels (SS304, SS304L, SS316, and SS316L). Some copper based metal alloys were reported by Farkas and Birchenall [89]. It is known that the copper is with a melting point of as high as 1083 °C. By synthesizing the metal alloys with aluminum, magnesium and so on, the melting temperature could be reduced to 400–500 °C [90].

Unlike organic EPCMs, metal alloys are typically synthesized using an electric muffle furnace with protection gas, such as argon, to prevent oxidation during the alloying process. The alloying process requires high temperatures and external mechanical stimuli to produce a homogeneous alloy. The reported metal alloy-based EPCMs are listed in Table 2 and summarized in Fig. 4.

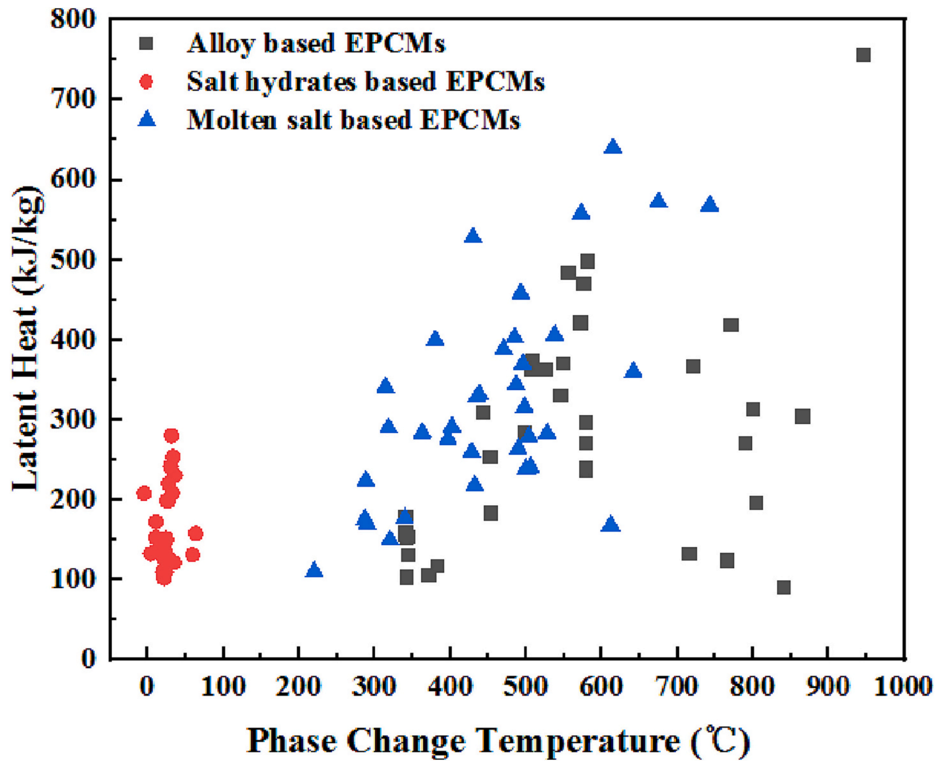

Fig. 4. Summary of inorganic EPCMs.

3.2.2. Salt hydrates based EPCMs

Compared to organic PCMs, hydrated salts are frequently used in medium and low temperature phase change energy storage materials due to their higher latent heat value, thermal conductivity, and wide range of phase change temperatures (ranging from 5 °C to 130 °C for various hydrated salts). Hydrated salts have the chemical formula $AB{\cdot}nH_2O$, with alkali metal salts as the main component and chloride, hydroxide, carbonate, nitrate, acetate, sulfate, phosphate, and other anions, along with water molecules, forming a new lattice composition. The complete phase transition process for single-component hydrated salts is $AB \bullet nH_2O + Q_{heat} \leftrightarrow AB + nH_2O$. However, incomplete melting of hydrated salts can occur due to the solubility of the salts. This incomplete phase transition process is $AB \bullet nH_2O + Q_{heat} \leftrightarrow AB \bullet (n-m)H_2O + mH_2O$. The phase transition process has a high latent heat value due to the breaking and formation of intermolecular bonds such as hydrogen bonds or inorganic chemical hydration bonds. Common hydrated salt PCMs include $CaCl_2{\cdot}6H_2O$, $Na_2SO_4{\cdot}10H_2O$, and $MgCl_2{\cdot}6H_2O$, which have a higher phase transition temperature. However, single-component hydrated salts can experience phase separation after phase transition and poor cyclability. EPCMs can mitigate phase separation during melting and solidification. Liu et al., [6] based on the solubility theory that inorganic salts can dissolve in another high-concentration salt solution [91], prepared an EPCM of $Na_2CO_3{\cdot}10H_2O$-$Na_2HPO_4{\cdot}12H_2O$ which does not undergo phase separation compared to single-component $Na_2CO_3{\cdot}10H_2O$.

Binary, ternary, and n-nary hydrated salt EPCMs can alleviate phase separation of single-component hydrated salts and have different effects on phase transition temperature and enthalpy. Li et al., [92] added different compositions of $MgCl_2{\cdot}6H_2O$ (10 wt%, 15 wt%, 20 wt%, 25 wt %) to $CaCl_2{\cdot}6H_2O$ and the phase transition temperatures were (26 °C, 25 °C, 23.5 °C, 21 °C), respectively. According to Table 2 (NO. 29-32), $CaCl_2{\cdot}6H_2O$ is mixed with Choline chloride, $MgCl_2{\cdot}6H_2O$, NH_4Cl and KCl to form binary or ternary EPCMs, the phase transition temperature is between 20 °C and 25 °C, making it suitable for building energy-saving applications. For hydrated salts with Na salts as their alkali metals, such as sodium carbonate decahydrate and sodium sulfate decahydrate, the phase transition temperature is between 25 °C to 35 °C, but the phase transition enthalpy can reach 280 kJ/kg. Liu et al., [93] prepared an EPCM of $Na_2SO_4{\cdot}10H_2O$ and $Na_2HPO_4{\cdot}12H_2O$ by melt blending. The EPCM not only avoids phase separation but also lowers the phase transition temperature to 31.2 °C with a large latent heat value of up to 280.1 kJ/kg. In Table 2 (NO. 36-41), at the same phase transition temperature, the hydrated salt n-nary EPCMs have a larger phase transition enthalpy than organic EPCMs, allowing them to store and release more latent heat. There are other nitrate hydrated salts, such as EPCMs formed by zinc nitrate and magnesium nitrate with ammonium nitrate [94], which have lower phase transition temperatures.

3.2.3. Molten salts

In addition to metals and alloys that have high phase transition temperatures and latent heat values, molten salt, an inorganic PCM, also exhibits high phase transition temperatures and latent heat values. The temperature range of molten salts used as EPCMs is typically within 300 °C ~ 1000 °C. Molten salts possess high stability, no phase separation, and low supercooling degree, making them suitable candidates for various industrial TES systems, particularly in concentrated solar thermal storage.

Molten salts can be classified into chloride salts, carbonate salts, nitrate salts, and fluoride salts. EPCMs can be prepared using a hybrid approach to alter the phase transition temperature, reduce corrosion, and mitigate issues such as thermal decomposition or supercooling. For example, in a study by Liu et al., [95] a molten salt EPCM was prepared using NaCl-Na_2CO_3. Compared to single-component chloride molten salts, which have a high corrosion problem [96], the addition of carbonate can improve thermal stability. Experimental results indicated that this EPCM exhibited excellent thermal stability without degradation after 1000 cycles between 600 and 650 °C. Additionally, the corrosion resistance of the material to stainless steel reached a stable value of 70 mg/cm^2 with an increasing number of cycles. Li et al., [97] prepared an 88 wt% $LiNO_3$-12 wt% NaCl EPCM by adding NaCl to

Table 2
Inorganic eutectic phase change materials.

Material number	Phase change material	Composition (wt%)	Phase change temperature (°C)	Heat of fusion (kJ/kg)	Reference
Alloy based EPCMs					
1	Mg:Zn:Al	70:24.9:5.1	339.9	157	[87]
		6:85.8:8.2	341.1	104	
		70:24.9:5.1	340	159.6	[88]
		7.3:84:8.7	343.7	131.5	
		35:6:59	443	310	[89]
2	Mg:Zn	28:72	341.1	152.7	[88]
		49:51	342	155	[99]
		48:52	340	180	[89]
		7.8:92.2	370.4	106.4	[88]
3	Al:Zn	11.3:88.7	381.7	118.4	
4	Al:Si	88:12	579	298	[100]
		83:17	578	272	
		80:20	578	237	
		70:30	578	241	
		87.8:12.2	580	499.2	[101]
5	Al:Mg	65.4:34.6	497	285	[102]
6	Al:Cu:Mg	60.8:33.2:6	506	364	
7	Al:Cu:Si:Mg	64.6:28:5.2:2.2	507	374	
8	Al:Cu:Si	68.5:26.5:5	525	364	
9	Al:Cu:Sb	64.3:34:1.7	545	331	
10	Al:Cu	66.9:33.1	548	372	
11	Al:Si:Mg	83.1:11.7:5.2	555	485	
12	Al:Si:Sb	86.4:9.6:4.2	575	471	
13	Al:Mg:Zn	59:33:6	443	310	[89,103]
14	Zn:Cu:Mg	15:25:60	452	254	
15	Al:Cu:Si	65:30:5	571	422	
16	Mg:Cu:Ca	52:25:23	453	184	
17	Cu:P	91:9	715	134	
18	Cu:Zn:P	69:17:14	720	368	
19	Cu:Zn:Si	74:19:7	765	125	
20	Cu:Si:Mg	56:27:17	770	420	
21	Mg:Ca	84:16	790	272	
22	Mg:Si:Zn	47:38:15	800	314	
23	Cu:Si	80:20	803	197	
24	Cu:P:Si	83:10:7	840	92	
25	Si:Mg:Ca	49:30:21	865	305	
26	Si:Mg	56:44	946	757	
27	Fe:Si:B	64.3:26.4:9.3	1157	1250 kWh/m^3	[104]
Salt hydrates based EPCMs					
28	$K_2HPO_4·3H_2O:NaH_2PO_4·2H_2O:Na_2S_2SO_3·5H_2O:H_2O$	6:6:6:82	−5.02	208.3	[105]
29	$CaCl_2·6H_2O$:Choline chloride	86:14	20.7	127.2	[106]
		89:11	23	135.2	
30	$CaCl_2·6H_2O:MgCl_2·6H_2O$	75:25	21.4	102.3	[107]
		85:15	23.9	151.9	[108]
31	$CaCl_2·6H_2O:NH_4Cl:KCl$	92.6:4.7:2.7	24.2	149.6	[109]
32	$CaCl_2·6H_2O:Fe_3Cl_2·6H_2O$	70:30	32	208	[110]
33	$LiNO_3·3H_2O:NaNO_3:Mn(NO_3)_2·6H_2O$	24.2:3:72.8	10.8	172.5	[111]
34	$LiNO_3·3H_2O:Mn(NO_3)_2·6H_2O:Mg(NO_3)_2·6H_2O$	22.9:68.6:8.5	10.8	152.8	
35	$NaNO_3:Ca(NO_3)_2: H_2O$	9.5:62.9:27.6	34.6	121.9	[112]
36	$Na_2SO_4·10H_2O:KCl$	95:5	25.3	128.7	[113]
37	$Na_2CO_3·10H_2O:Na_2HPO_4·12H_2O$	50:50	26.9	198.4	[114]
		40:60	27.3	220.2	[7]
		42:58	25.0	199.1	[115]
38	$Na_2SO_4·10H_2O:Na_2CO_3·10H_2O$	50:50	23.9	110.3	[116]
39	$Na_2CO_3·10H_2O:MgSO_4·7H_2O$	70:30	35.6	230.5	[117]
40	$Na_2SO_4·10H_2O:Na_2HPO_4·12H_2O$	80:20	33.4	253.5	[118]
		25:75	31.2	280.1	[119]
41	$Na_2SO_4·10H_2O:Zn(NO_3)_2·6H_2O$	70:30	30	242	[120]
42	$MgCl_2·6H_2O:Mg(NO_3)_2·6H_2O$	41.3:58.7	59.2	131.3	[121]
43	$MgCl_2·6H_2O:NH_4Al(SO_4)_2·12H_2O$	30:70	63.4	157.8	[122]
44	$Zn(NO_3)_2·6H_2O:Mn(NO_3)_2·4H_2O:KNO_3$	33.3:33.3:33.4	18–21	110	[123]
45	$Zn(NO_3)_2·6H_2O: NH_4NO_3$:	75:25	12.4	135	[94]
46	$Mn(NO_3)_2·6H_2O: NH_4NO_3$:	73:27	3.9	133	
Molten salt based EPCMs					
47	$LiCl:LiCrO_4:LiVO_3:LiF$	41.5:35.1:16.4:7.0	340	177	[124]
48	$LiCl:LiVO_3:LiF:Li_2SO_4:Li_2MoO_4$	42.0:17.4:16.2:11.6:11.6	363	284	[125]
49	$LiCl:LiF:Li_2SO_4:Li_2MoO_4$	51.5:16.2:16.2:16.2	402	291	[126]
50	LiOH:LiF	80:20	427	1163	[127]
		20:80	430	528	[128]
51	$LiVO_3:LiF:Li_2MoO_4:Li_2SO_4$	43.8:25:16.5:14.8	428	260	[129]

(continued on next page)

Table 2 (*continued*)

Material number	Phase change material	Composition (wt%)	Phase change temperature (°C)	Heat of fusion (kJ/kg)	Reference
52	$LiF:KF:NaF:BaF_2$	45.7:41.2:11.3:1.8	438	332	[128]
53	LiF:LiCl	73.6:26.4	485	403	[128]
54	LiCl:KF	50:50	487	344	[128]
55	LiF:KF	33:67	493	458	[130]
56	$Li_2CO_3:Na_2CO_3$	44:56	496	370	[131]
57	$Li_2CO_3:K_2CO_3$	28.5:71.5	498	316	[132]
58	$Li_2SO_4:Li_2MoO_4:CaMoO_4$	59.8:36.7:3.5	538	406	[133]
59	$LiF:NaF:CaF_2$	52:35:13	615	640	[127]
60	$NaOH:NaCl:Na_2\ CO_3$	77.2:16.2:6.6	318	290	[134]
61	$NaF:NaNO_3:NaCl$	5:87:8	288	224	[84]
62	NaF:NaBr	27:73	642	360	
63	NaF:NaCl	33.5:66.5	675	572	
64	$NaF:CaF_2:MgF_2$	65:23:12	743	568	
65	$NaCl:MgCl_2$	61.5:38.5	435	328	[135]
66	$Na_2MoO_4:NaBr:NaF$	55:43:2	506	241	[136]
67	$NaCl:Na_2MoO_4:NaBr$	38.5:38.5:23	612	168	[137]
68	$NaCl:NiCl_2$	52:48	573	558	[127]
69	$NaNO_3:KNO_3$	90:10	290	170	[138]
		60:40	220	110	
70	$Na_2SO_4:NaCl:NaNO_3$	8.8:5.7:85.5:	287	176	
71	KOH:LiOH	60:40	314	341	[139]
72	$K_2CO_3:Na_2CO_3:Li_2CO_3$	34.5:33.4:32.1	397	276	[139]
73	$KCl:K_2CO_3:KF$	40:37:23	528	283	[131]
74	$KNO_3:KCl$	95.5:4.5	320	150	[138]
75	$KCl:ZnCl_2$	54:46	432	218	
76	$MgCl_2:KCl:NaCl$	60:20.4:19.6	380	400	[139]
77	$MgCl_2:KCl$	64:36	470	388	[84]
78	$CaCl_2:CaF_2:NaF$	50:1.5:48.5	490	264	
79	$CaCl_2:NaCl$	52.8:47.2	500	239	
80	$CaCl_2:KCl:NaCl$	66:5:29	504	279	

$LiNO_3$. The addition of NaCl decreased the supercooling degree of the EPCM to as low as 1.2 °C, and the decomposition temperature of the mixed molten salt increased from 560 °C to 620 °C, significantly increasing the applicable temperature range of $LiNO_3$. According to Table 2, the phase transition temperature of eutectic nitrate PCM is lower than that of chloride and carbonate. Compared to other molten salts, such as carbonates with high liquid viscosity, some carbonates that are easily decomposed, chloride salts with greater corrosiveness, and fluoride salts that have noticeable disadvantages during the solid-liquid phase change process, nitrate salt-based EPCMs have the advantages of being low-cost, having low corrosion, and not decomposing at 500 °C, but exhibit poor thermal conductivity and local overheating problems. Another way of classifying molten salt cations is by dividing them into alkali metal salts, such as lithium salts (NO. 47-59), sodium salts (NO. 60-70), and potassium salts (NO. 71-77). However, lithium salts and silver salts are not widely employed due to their high cost. According to [98], sodium salt and potassium salt-based EPCMs have melting temperatures between 200 °C to 700 °C.

The inorganic EPCMs listed in Table 2 are summarized in Fig. 4. Hydrated salt-based EPCMs have a phase transition temperature below 50 °C and a latent heat value between 100 kJ/kg to 300 kJ/kg. On the other hand, molten salt-based and metal-based EPCMs have high phase transition temperatures and latent heat values. Some metal-based EPCMs have a phase transition temperature above 700 °C, making them suitable for high-temperature phase change energy storage applications.

3.3. Organic-inorganic EPCMs

In addition to binary, ternary or *n*-nary EPCMs formed by organic-organic, inorganic-inorganic PCMs or organic-inorganic EPCMs have also been proposed as eutectic phase change energy storage materials. For instance, Li et al., [140] proposed a binary EPCM by fusing magnesium nitrate hexahydrate and glutaric acid. When the mass fraction is 60:40, the eutectic temperature reaches 66.7 °C, and the latent heat value is 189 kJ/kg. The study mentioned that it was possible to use the eutectic to change the crystal structure to increase the latent heat. Compared with the original latent heat value of 165.2 kJ/kg, the latent heat value increased by 14.4 %. Moreover, Wang et al., [141] prepared a 70 wt% $Na_2HPO_4{\cdot}12H_2O$ (DHPD):30 wt% CA EPCM. The addition of CA reduces the degree of supercooling of DHPD and the degree of supercooling of EPCM is reduced to 0.9 °C. Additionally, the addition of DHPD increases the latent heat of phase change of CA to 168.8 kJ/kg and the thermal conductivity of the EPCM is 0.468 W/(m·K). Interestingly, the prepared organic-inorganic EPCMs have the characteristics of increasing heat storage density and providing a direction for the preparation of high-density heat storage materials in the future. Sun et al., [142] also prepared a glycerin/sodium acetate trihydrate (SAT) based inorganic-organic EPCM to reduce the viscosity of glycerin and improve thermal conductivity, when the preparation ratio is 2:1, the viscosity of the system is reduced by 58.03 %, and the thermal conductivity is increased by 17.2 %.

4. Limitations and performance enhancement

The low thermal conductivity and poor cyclic thermal stability are the primary factors hindering the large-scale application of EPCMs. Phase separation can alter the composition of the PCMs, causing fluctuations in temperature and latent heat value during repeatable cycles. In addition, supercooling during liquid solidification can impede heat release, limiting practical applications. This section will discuss techniques aimed at improving the thermal conductivity, phase separation, and supercooling of EPCMs.

4.1. Thermal conductivity enhancement

In light of the issues of low thermal conductivity and poor heat transfer performance of EPCMs, this section provides an overview of the major methods proposed in recent years to improve the thermal conductivity of EPCMs. There are three main approaches to enhance the heat transfer performance of PCMs, including the addition of extended heat transfer surfaces in heat exchangers, the addition of high thermal conductivity additives into PCMs, and the use of high thermally conductive porous materials to form a skeleton structure. For instance, Hafiz Muhammad Ali [143] proposed the use of PCM foamed copper and heat pipes together for radiator cooling technology, and the results demonstrated that the addition of foamed copper increased the heat dissipation rate by two times compared to non-added copper foam. Additionally, nickel foam [144] can also be utilized as a high thermal conductivity material to improve the thermal conductivity of heat dissipation equipment. In order to maintain the high heat storage density of EPCMs, high thermal conductivity agents or high porosity thermal conductivity materials are typically added to EPCMs to prepare composite EPCMs, as illustrated in Fig. 5. Carbon and metal nanoparticles are commonly used as additives. Carbon-based additives are widely used due to their high thermal conductivity and reliable stability. Carbon-based thermal conductivity enhancement additives typically include expanded graphite, carbon nanotubes, carbon fibers, and other carbon-based materials. Detailed review articles describing the advances in thermal conductivity enhancement for single-component PCM are available. For example, Tao et al., [145] have summarized the use of porous media and nano-additives as the current mainstream methods to improve the thermal conductivity of PCMs. In this context, Hafiz Muhammad Ali et al., [146] also proposed expanded graphite, encapsulated PCMs, hexagonal boron nitride, and spongy graphene to improve the thermal conductivity. This section mainly focuses on the thermal conductivity enhancement techniques for EPCMs.

Based on the findings presented in Table 4, the low thermal conductivity of EPCMs can be significantly improved by filling them with expanded graphite (EG). For instance, Zhang et al., [71] prepared a PA-SA binary EPCM with varying amounts of EG by stirring and baking. When the mass fraction of PA-SA:EG was 13:1, the thermal conductivity of the EPCM filled with EG increased from 0.26 W/(m·K) to 2.51 W/(m·K). Similarly, Zhang et al., [40] improved the thermal conductivity of a CA-PA-SA ternary EPCM by filling it with 10 % EG, as shown in Fig. 6(a). The thermal conductivity increased linearly with continued compression of the EPCM, as illustrated in Fig. 6(b). The use of EG as a filler also enhanced the thermal conductivity of inorganic-inorganic EPCMs. Couto et al., [53] prepared multi-component PCMs consisting of eutectic nitrates and EG using various elaboration methods such as infiltration, hot and cold compression. They achieved at least a five-fold increase in effective thermal conductivity. Huang et al., [147] developed a molten salt $LiNO_3$-KCl and EG for high-temperature solar thermal storage, where the thermal conductivity of the EPCMs was found to be 1.85 to 6.65 times higher than that of the pure EPCMs for EG contents between 10 % to 30 %. Lopez et al., [148,149] created KNO_3-$NaNO_3$/

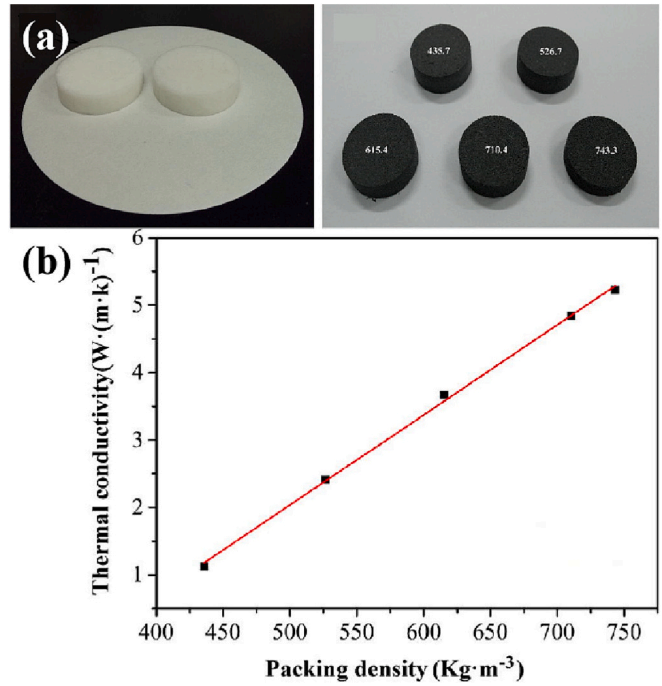

Fig. 6. Filled EG improves thermal conductivity of EPCMs, adapted from Ref. [40].

graphite composites through simple uni-axial and isostatic compression of salt powders and expanded natural graphite particles. The resulting EPCMs with graphite content between 15 wt% and 20 wt% exhibited a thermal conductivity of up to 20 W/(m·K). This section provides an overview of the thermal conductivity enhancement techniques for EPCMs, with a focus on the use of EG as a filler.

The use of carbon fiber or carbon nanotubes (CNTs) additives is mainly limited to low-temperature energy storage applications, while EG has proven successful in high-temperature molten salt eutectic energy storage. For instance, in their study on improving the thermal conductivity of PA-SA, Zhang et al., [150] added CNTs to the EPCM and observed a thermal conductivity increase of 20.2 %, 26.2 %, 26.2 % and 29.7 % when the mass fraction of CNTs was 5 wt%, 6 wt%, 7 wt%, and 8 wt%, respectively. It is worth noting that the CNT-filled EPCMs retained

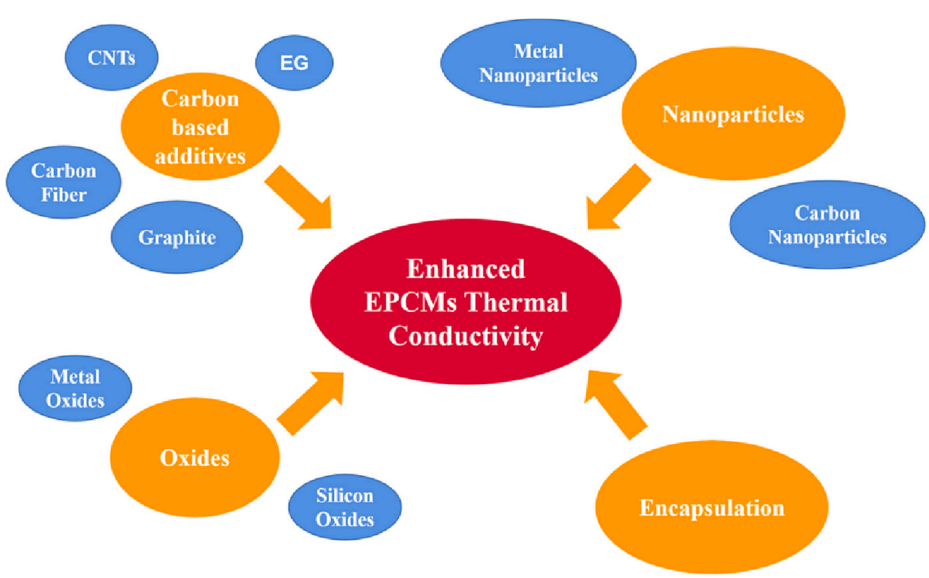

Fig. 5. Measures to enhance EPCMs thermal conductivity.

good thermal conductivity even after 600 cycles. Apart from carbon-based additives, metal oxides or micro/nano encapsulation are also employed for thermal conductivity enhancement. For example, Liu et al., [119] dispersed α-Al_2O_3 nanoparticles into $Na_2SO_4·10H_2O$-$Na_2HPO_4·12H_2O$ binary EPCMs, which improved their thermal conductivity. At the liquid state, the addition of 3 wt% and 4.5 wt% α-Al_2O_3 nanoparticles improved the thermal conductivity by 10.3 % and 21.7 %, respectively, compared to the unfilled EPCM. At the solid state, the thermal conductivity increased by 26.8 % and 61.3 %, respectively. Imra et al., [151] prepared nanoparticle-modified EPCMs by using OA-polyethylene glycol (PEG) as the core binary EPCM and SiO_2/SnO_2 as the shell material prepared by hydrolysis, polycondensation, and silanization at the in-situ emulsion interface. Baskar et al., [152] reported a novel EPCM prepared using LA, PA and SiO_2, the thermal conductivity was improved by 54.4 % with 5 % addition of SiO_2 compared to LA-PA eutectic. Among these methods, silica shell material and silica/tin dioxide shell are used to improve the thermal conductivity, with the original thermal conductivity of 0.2497 W/(m·K) increasing to 0.5538 W/(m·K) and 0.7053 W/(m·K), respectively.

While the method of adding high thermal conductivity agents has demonstrated good thermal conductivity improvement and cyclic stability, investigations on the skeleton structure in EPCM and extended surface in heat exchangers are rarely reported for EPCM. The most commonly used additive is EG, which not only improves the thermal conductivity of EPCM but also prevents leakage of the EPCMs due to its porous properties. EG is relatively inexpensive and can withstand molten salt corrosion at high temperatures compared to metal oxides and metal nanoparticles.

4.2. Supercooling suppression

The supercooling phenomenon typically occurs during the solidification process of PCMs. In the case of EPCMs, when the temperature falls below the phase transition temperature, the liquid does not spontaneously change its phase to solidify, resulting in supercooling. The degree of supercooling is defined as the temperature difference between the theoretical and actual solidification temperature. Solidification occurs when the embryo radius grows larger than the critical nucleation radius, as illustrated in Fig. 7. To improve the efficiency of heat storage and release and reduce the loss of latent heat value, it is necessary to reduce the degree of supercooling of EPCMs. The main factors that affect the degree of supercooling are cooling rate, sample purity, water content, melting time, and melting temperature, as depicted in Fig. 8. Fig. 8 shows that there are mainly two ways to promote crystallization in a liquid solution: adding seeding and dynamic nucleation. Unlike adding nucleating agents to EPCMs to promote crystallization, dynamic nucleation utilizes external fields such as agitation, stirring, cold finger, and ultrasonic vibration to trigger crystallization. Most research for reducing the degree of supercooling in EPCMs employs the addition of nucleating agents. This section summarizes previous works that employed the addition method to suppress the supercooling effect.

The lattice structure of the nucleating agent should not differ from that of the EPCM by >15 % to maintain effective nucleation promotion. Nanostructured nucleating agents, such as nanoparticles, are highly effective in inducing solution crystallization. Not only can they reduce the degree of supercooling, but they can also improve the thermal conductivity of the EPCM. Multi-layer CNTs are frequently employed as nucleating agents in research due to their exceptional mechanical and thermal properties, as well as their resistance to corrosion and nanoscale effects. Multi-walled carbon nanotubes (MWCNTs) have been added to Na_2CO_3-MgO composite PCMs by Ye et al., [153], which resulted in significant improvements in thermal conductivity and thermal cycling stability. Similarly, in organic-organic EPCMs, the addition of MWCNTs has been shown to improve thermal conductivity and reduce supercooling. For example, Hussain et al., [154] added MWCNTs to CA-LA and reduced the supercooling degree from 0.62 °C to 0.37 °C, while increasing the thermal conductivity from 0.269 W/(m·K) to 0.397 W/(m·K). Surfactants [65] and sodium tetraborate decahydrate [113] have also been employed to reduce the degree of supercooling in solutions. In addition to being an effective high thermal conductivity material, EG has been used as a supercooling reduction material in some eutectic systems. For instance, Ling et al., [155] added EG to the

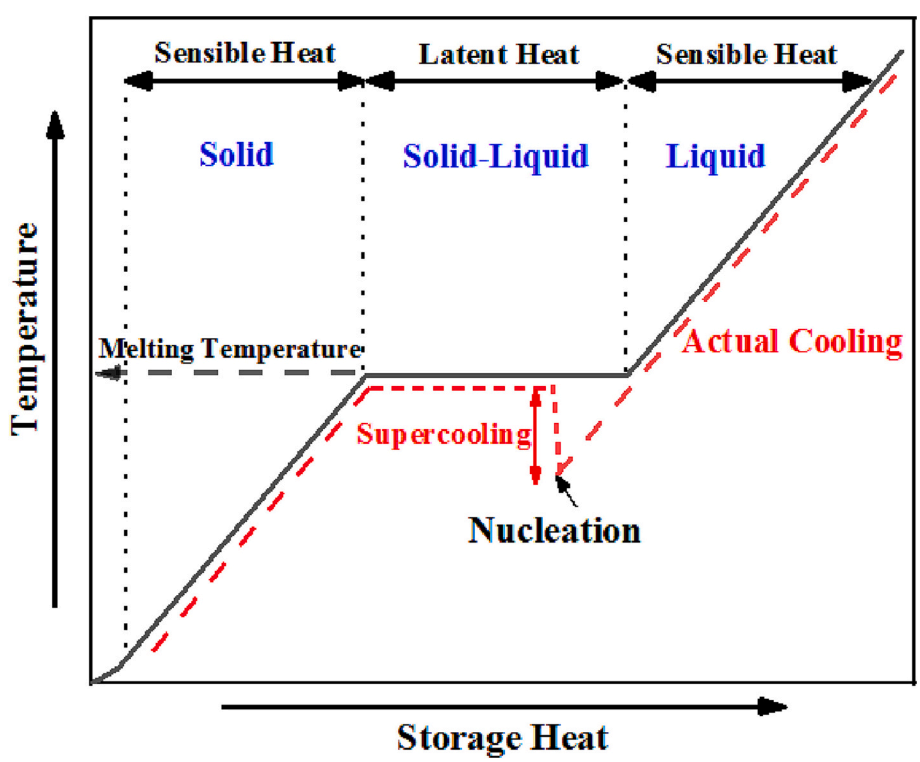

Fig. 7. Phase transition process of EPCM with certain degree of supercooling.

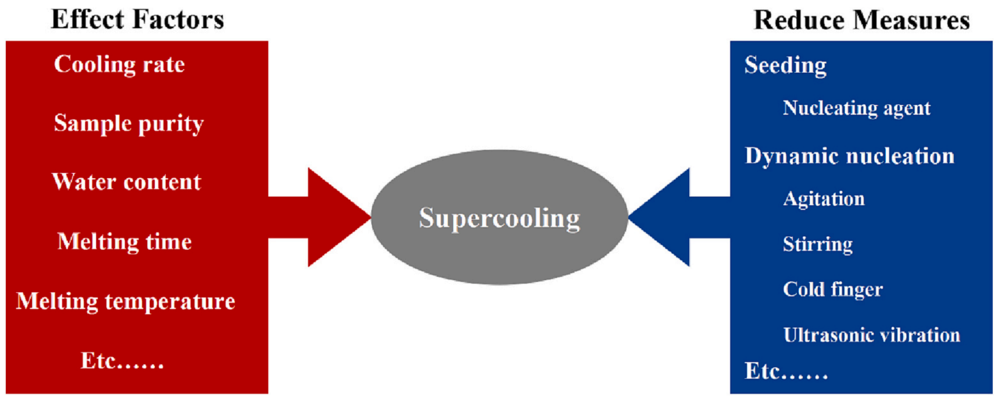

Fig. 8. Influencing factors of supercooling and reducing measures.

$MgCl_2{\cdot}6H_2O$–$Mg(NO_3)_2{\cdot}6H_2O$ binary system and reduced the degree of supercooling from 18.4 °C to 2.7 °C. In the case of multiple eutectic systems, the addition of EG to the $K_2HPO_4{\cdot}3H_2O$–$NaH_2PO_4{\cdot}2H_2O$–$Na_2S_2O_3{\cdot}5H_2O$–H_2O eutectic system was shown to reduce supercooling to 1.8 °C [105].

4.3. Phase separation inhibition

In inorganic PCMs, the phenomenon of phase separation typically occurs after repeated cycles due to the difference in salt solubility. This phenomenon accumulates with each cycle, eventually leading to a change in the composition of the EPCMs, and also adversely affecting the phase transition temperature as well as the latent heat value. The phase separation of the hydrated salt is mainly due to the fact that after the hydrated salt loses crystal water, the density of the salt particles becomes greater than that of the solution. As a result, the salt particles in the solution are precipitated at the bottom, forming a heterogeneous mixture in the EPCM solution. When solidification occurs again, only a part of the salt particles at the bottom recombine with the crystal water to form hydrated salts, resulting in only a partial crystallization of hydrated salts in the solution, while the rest mainly consist of salt particles and excess water [156].

To minimize the adverse effect of phase separation, three major methods are employed, as shown in Table 3. The first method involves filling the PCM with additional water to alter the solubility of hydrated salts. However, this can reduce the latent heat value or heat storage density of the EPCMs. The second method involves controlling the diffusion rate by dividing the EPCMs into several barriers. By reducing the size of the container, the diffusion rate is inversely proportional to the square of the size, and thus, the phase separation of the EPCMs can be effectively alleviated. However, this method is not suitable for large-scale application of EPCMs as it requires a significant number of barriers. It is noteworthy that adding excess water or reducing the size are not suitable for large-scale application of EPCMs [157]. The third and increasingly effective method is to add thickeners to promote the thickening of the PCMs. For instance, Hou et al., [113] added different mass fractions of potassium chloride into sodium sulfate decahydrate to prepare an EPCM. To reduce the phase separation of the solution, they added 5 wt% polyacrylamide as a thickening agent, 5 wt% sodium tetraborate as a nucleating agent and EG as a high thermal conductivity medium, which significantly reduced the phase separation and supercooling degree to 0.4 °C. Other studies have also added EG to EPCMs to reduce phase separation [105,155], EG can not only be used as a high thermal conductivity material additive, but also a porous material which can make it more uniformly dispersed by increasing the viscosity of the hydrated salt solution, and fix the crystallization water of the hydrated salt inside by microporous adsorption, avoiding phase separation [155]. Expanded vermiculite [158] and the like are also used for the same purpose. Expanded vermiculite and other similar materials have also been used for this purpose. According to the literature [159], after adding CNTs to magnesium chloride eutectic solution, no phase separation was observed even after 400 cycles. Furthermore, several studies have shown that the addition of EG not only improves thermal conductivity but also promotes the reduction of supercooling and phase separation [160].

Table 3
Measures to inhibit phase separation.

Measurements	Advantage	Disadvantage
Water addition	Simple, convenient, wide source	Severely change phase change temperature, change latent heat, increase supercooling
Size reduction	Well compatibility, less impact on phase change temperature, less impact on latent heat	Complex operation, high cost, not suitable for engineering applications
Thickeners addition	Sample operation, cycle stability, little effect on EPCMS	Moderate price, increase solution viscosity

Summarizing the research works published in recent years for improving the performance of the EPCMs as shown in Fig. 9, the main limitations of the application of EPCMs are low thermal conductivity, large degree of supercooling and phase separation. The improvement measures mainly include adding additives with high thermal conductivity, such as EG, CNTs, metal-based oxides and other additives, additives with nucleating and thickening agents to reduce subcooling and reduce phase separation, respectively. The improvement methods in thermal conductivity increment, supercooling degree reduction and phase separation inhibition are summarized in Table 4. Besides, the performance comparison before and after improvement is drawn.

In addition to the above-mentioned factors limiting EPCMs, such as low thermal conductivity, large degree of supercooling, and phase separation, it is severely corrosive to high-temperature EPCMs like molten salts [95] and alloys [161] and low-temperature EPCMs, such as fatty acids EPCM [162,163] and hydrate salt [156]. The corrosion resistance of PCMs has been reviewed [164], such as adding corrosion inhibitors, encapsulating PCM or anti-corrosion coatings.

5. Applications

EPCM is divided into low temperature (lower 25 °C), medium temperature (25–300 °C) and high temperature (exceed 300 °C) thermal storage environment according to the application, which have more advantages of wide phase transition temperature range, large latent heat value and high heat storage density as compared with single-component PCMs. This section summarizes the applications of EPCMs in different fields including cold storage, building thermal comfort, biomedical science, PV/T systems, power battery thermal management and solar-thermal power plants.

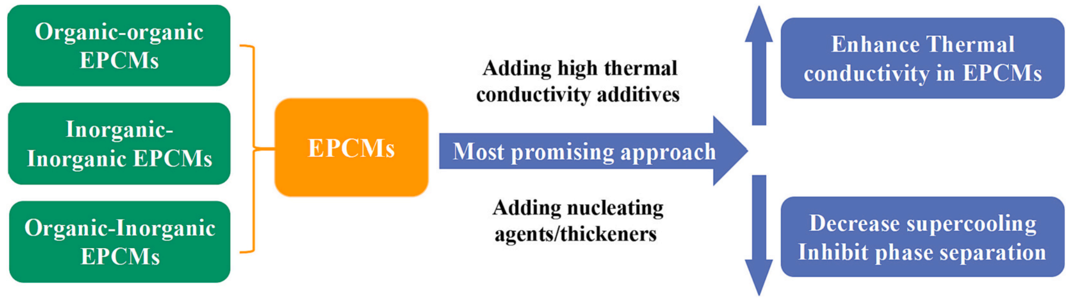

Fig. 9. Summary of the way to improve EPCMs in the future.

5.1. EPCM for cold storage

Due to its wide phase transition temperature, EPCMs can not only store high to medium temperature thermal energy, but also could store low temperature thermal energy. For household refrigerators, when the refrigerator is running with peak-valley price, the cold storage at valley price and release at peak price would bring benefit to the customers.

Abdolmaleki et al., [172] used PEG with different molecular numbers to prepare EPCM with PEG200 and PEG300 as raw materials. When 30 wt% PEG200-70 wt% PEG300, the eutectic temperature was −20 °C, as shown in Fig. 10, the room temperature and energy consumption tests in the refrigeration system with and without the addition of EPCM were tested. The experimental results show that the application of EPCM in the freezer can significantly reduce the temperature fluctuation by 40.59 %, the optimum value of polyethylene glycol is 2 kg, and the energy consumption of the freezer using EPCM is significantly lower compared with ordinary refrigerators, the energy saving is 8.37 % at 1.5 kg PCM. In addition to the application of organic EPCMs in the field of low temperature, Gin et al., [173] also studied the application of EPCM in the defrosting of refrigerators and the energy saving of opening and closing doors, and prepared an NH_4Cl solution with a eutectic temperature of −15.4 °C. When the refrigerator is not working, EPCM can be used as a cooling source. Experimental results showed that adding EPCM to the freezer reduced energy consumption by 8 % during the defrost cycle and by 7 % during the system's door opening period. Jonathan et al., [174] also studied the performance improvement of low-temperature EPCMs used in refrigerators, and selected two EPCMs, one is commercial PLUSICE E-10 and the other is 19.5 wt% NH_4Cl solution. The two EPCMs are made to reduce the start-stop frequency of the compressor and increase the evaporation temperature. The packages containing the two EPCMs are placed in the cabinet of a commercial refrigerator. The temperature at different positions in the refrigerator is tested. The phase transition temperatures of the two PCMs are −11.2 °C and −15.6 °C. Finally, the compressor start-stop time was reduced by 3.7 % and 9 % with the addition of two EPCMs. The energy consumption of the evaporator was reduced by 1.74 % and 5.81 %.

EPCMs can not only be used as a cold source for cold storage in household refrigerators, but also can be used as a cold source in other occasions such as food or biological cold chain, Zhao et al., [175] prepared a EPCM suitable for the temperature requirements of pharmaceutical cold chain logistics, and developed a EPCM with a phase change temperature of 2 °C ~ 8 °C. With the eutectic salt of 66 wt% tetradecane-34 wt% lauryl alcohol, the eutectic temperature is 4.3 °C, and the latent heat value is 247.1 kJ/kg. In order to improve the thermal conductivity, EG is added, and the thermal conductivity is increased by 3.53 times. The developed EPCM is coupled with the refrigeration equipment for experiments. The results showed that the average cold keeping time in the vaccine refrigerator at 2 °C ~ 8 °C was 44.37 h when the load was empty, and the average holding time during the experiment was 46.04 h when the load was added. Liu et al., [176] propose a novel EPCM capable of meeting the temperature range (2–8 °C) of vaccine cold chain logistics, decyl alcohol and lauric acid were selected as co-crystals. The results show that the phase transition temperature of DA-LA/EG EPCM is 2.08 °C and the latent heat is 188.71 kJ/kg. Taking yogurt as an object, when the ambient temperature is 3 °C, the cold storage time of the incubator filled with EPCM can be close to 10 h. The temperature field of the incubator is uniform, and the temperature of each plate is the same. After 10 h, the pH and activity of the probiotics in the yogurt remained stable.

5.2. EPCM for building thermal comfort

According to the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), the temperature range for indoor thermal comfort is 19 °C to 29 °C [177]. The indoor comfort in a building refers to the most suitable temperature in the building and the suitable temperature range is relatively small. Traditional single-component PCMs cannot meet the requirements for indoor comfort due to limited phase transition temperature range. Since the binary and *n*-nary EPCMs extends the phase transition temperature as compared to the single-component PCMs, EPCMs are widely employed for maintaining indoor thermal comfort [178]. When the indoor temperature increases/decreases, the EPCMs melts/solidifies and undergoes a phase change absorbs/releases heat and then maintains a suitable and stable temperature in the room.

The first thing to consider when EPCMs are applied in the construction field is corrosion, and the corrosion of construction metals has received more and more attention. Veerakumar et al., [51] prepared a binary EPCM of 40 wt% lauric acid-60 wt% myristyl alcohol. The phase transition temperature is 21.3 °C and the latent heat is 151.5 kJ/kg. After 1000 cycles, it still remains thermally stable as shown in Fig. 11, which shows that EPCM is compatible with Al and SS316, but shows corrosion with Cu. The experiment result indicating that the prepared EPCM is a good candidate for improving building thermal comfort. Chinnasamy Veerakumar et al., [179] proposed a capric acid/cetyl alcohol EPCM for building thermal comfort. The results showed when its composition is 70 wt%:30 wt%, the phase transition temperature is 22.89 °C, and the latent heat value is 144.92 kJ/kg, after 1000 cycles the EPCM was found without significant change and EPCM also have good compatibility with aluminum and stainless steel 316 except copper. In addition to the compatibility of EPCMs and building materials, Ye et al., [180] prepared binary eutectic hydrated salts of $CaCl_2·6H_2O$ (CCH) and $Mg(NO_3)_2·6H_2O$ (MNH), when the amount of MNH added is 6 wt%, the phase transition temperature of the eutectic material is 27.51 °C, and the latent heat value is 142.4 kJ/kg. In order to prevent the leakage of EPCM liquid, continue to add silica to prepare a EPCM. The experimental results show that compared with the reference room, when the EPCM thickness is 30 mm, the energy consumption can be saved by 190.05 kWh in summer and 361.46 kWh in winter, and the addition of EPCM can significantly reduce the indoor temperature fluctuation. Hydrated salt EPCMs are rarely used in building energy storage due to leakage and corrosion problems, and most of the literature is still dominated by organic PCMs. Baskar et al., [152] proposed to develop a new EPCM using nano-silica (SiO_2) combined with LA and PA eutectic mixture to

Table 4
Improved performance of eutectic phase change materials.

EPCMs	Thermal properties			Improvement measures	Improved performance parameters			Reference
	Thermal conductivity (W/m·K)	Supercooling (°C)	Phase separation		Thermal conductivity (W/m·K)	Supercooling (°C)	Phase separation	
CA-MA	0.15	–	–	Vacuum impregnation method and added EG to prepare CA-MA/VMT/EG	0.22	–	–	[165]
CA-MA-PMMA	0.269	–	–	Bulk- polymerization method to prepare CA-MA/ PMMA	0.509	–	–	[43]
PA-SA	0.263	–	–	8 wt% CNTs was added to PA-SA/CNTs	0.341	–	–	[150]
CA-MA-PA	0.149	–	–	Ultrasonic method using xG-F as carrier to prepare CA-MA-PA/xG-F	0.170	–	–	[166]
PA-LA	0.24	–	–	Vacuum impregnation method PA-LA/ carbonized abandoned rice	0.44	–	–	[167]
ES-EP	0.257	–	–	Nanocapsules of ES-EP eutectic mixture coated with metal	0.713	–	–	[168]
LA-MA-PA	0.21	–	–	EG absorbed by LA-MA-PA	1.67	–	–	[53]
PA-SA	0.26	–	–	Adding EG to prepare PA-SA/EG	2.51	–	–	[71]
SA-AA	0.12	–	–	Adding EG to prepare SA-AA	1.74	–	–	[169]
$LiNO_3$-KCl	1.749	–	–	Adding EG to prepare $LiNO_3$-KCl	5–15	–	–	[147]
$Na_2SO_4·10H_2O$-$Na_2HPO_4·12H_2O$	0.7892	7.8	–	Adding nano-α-Al_2O_3 to $Na_2SO_4·10H_2O$-$Na_2HPO_4·12H_2O$	1.273	1.6	–	[119]
OA-PEG	0.2497	0.63	–	Adding SiO_2/SnO_2 as shell then using successive ionic layer adsorption and reaction method to prepare nano encapsulated PCMs	0.7053	0.54	–	[151]
CA-LA	0.2692	0.62	–	Adding MWCNT to CA-LA	0.3972	0.37	–	[154]
AA-SA	0.11	17.7	–	Adding GNP to AA-SA	0.131	3.6	–	[170]
MA-PA	0.225	1.97	–	Adding Surfactant additives to MA-PA	0.242	0.34	–	[65]
m-Erythritol-D-mannitol	0.33	–	–	Adding EG to prepare m-Erythritol-D-mannitol	0.71	62.6	–	[29]
SA-BA	0.3396	2.74	–	Adding EG to prepare composite PCMs	4.177	1.3	–	[76]
MA-PA	0.225	1.97	–	Adding Sodium laurate (10 %) to MA-PA	0.235	0.43	–	[24]
Galactitol-mannitol	0.305	70.1	–	Vacuum impregnation method to prepare Galactitol-mannitol-EG	9.439	51.6	–	[171]
$CaCl_2·6H_2O$-$MgCl_2·6H_2O$	0.732	–	–	Adding SrCl2·6H2O to inhibite the supercooling, EPCM was formed by impregnation method with EP	0.144	Little supercooling	No phase separation	[108]
$MgCl_2·6H_2O$-$Mg(NO_3)_2·6H_2O$	0.413	5.13	Significant phase separation	Composited with porous fumed silica to prepare EPCM	0.434	2.91	No apparent phase separation	[121]
$K_2HPO_4·3H_2O$-$NaH_2PO_4·2H_2O$-$Na_2S_2O_3·5H_2O$-H_2O	0.669	5.4	Significant phase separation occurred once in the thermal cycle	Using impregnation method to add Al_2O_3-coated EG to prepare EPCM	8.9	1.83	No significant phase separation still occurred in 400 thermal cycles	[105]
$Na_2SO_4·10H_2O$-KCl	0.56	15.7	Severe phase separation occurred	Adding Polyacrylamide, sodium tetraborate decahydrate and EG was proposed as thickener, nucleating agent and the high thermal conductivit medium	1.35	2.3	Very little phase separation	[113]
$MgCl_2$-H_2O	0.525	16.562	–	Adding MWCNTs and nucleating agents	0.5344	2.167	No significant phase separation was observed after 400 thermal cycles	[159]

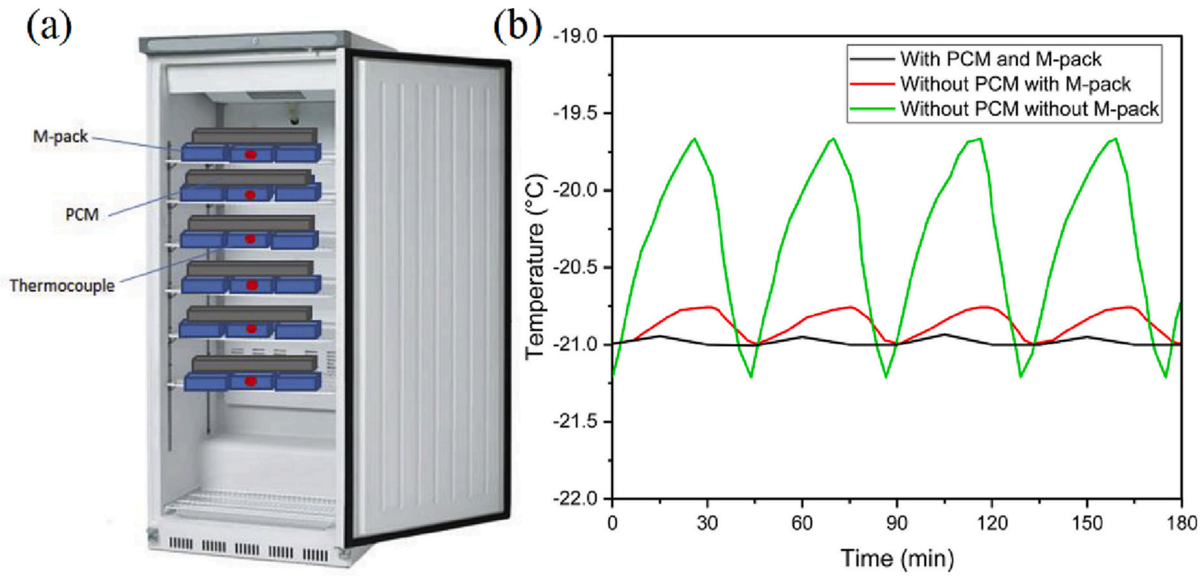

Fig. 10. Filled EPCMs for low temperature household cold storage, adapted from Ref. [172].

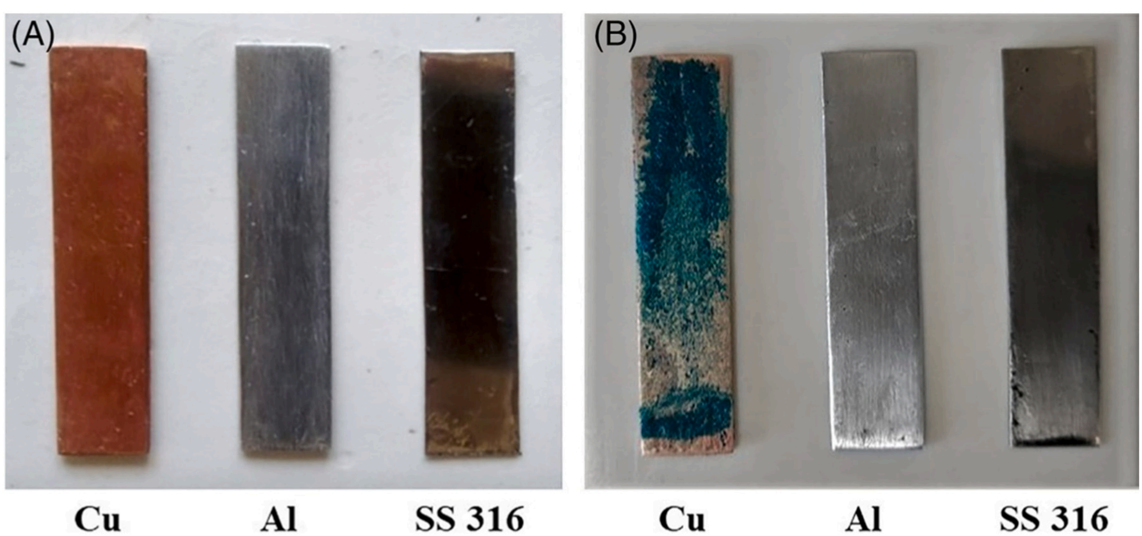

Fig. 11. EPCMs using in improving the building thermal comfort, adapted from Ref. [51].

improve thermal energy in buildings storage performance. The results confirmed that the melting range of the EPCM was 36 °C, the latent heat was as high as 187 kJ/kg, and the thermal conductivity value of the composite PCM containing 5 wt% SiO_2 was increased by 54.4 %. Therefore, the use of SiO_2-loaded composite PCMs in building materials can store a large amount of thermal energy and maintain thermal comfort to the greatest extent. Zhang et al., [181] in this study, a binary eutectic mixture of erythritol and urea was prepared and EG was used to reduce supercooling and improve the thermal conductivity of the binary material. The EPCM integrated solar heating system can save 5 tons of standard coal in one heating season for a building with a heating area of 1125 m^2 in Tianjin, China, compared to the conventional heating mode. Therefore, it is recommended as a promising PCM for solar energy storage.

5.3. EPCM for biomedical science

In recent years, organic PCMs can also be used in the field of biomedical sciences including body temperature maintaining and target medicine release. Fatty acid based organic PCMs are commonly employed due to their low cost, good chemical stability and biocompatibility. However, the phase transition temperature of the single-component fatty acid PCMs (Phase change temperature almost above 40 °C) is difficult to be applied for human body. To solve this problem, organic fatty acid EPCMs have been proposed to expand the phase change temperature range to be suitable for human body environment [182].

Zhu et al., [183] proposed a binary fatty acid EPCM with a melting temperature of 39 °C using LA:SA in a ratio of 4:1. The EPCM was employed as a near-infrared-triggered drug release gate material. After the dye absorbs near-infrared radiation (NIR) light, the temperature is increased according to the photothermal effect, the EPCM is melted. The melting of the EPCM helps the release of the drug as shown in Fig. 12. Behrouz et al., [184] prepared a binary EPCM LA:PA with 69 wt%:31 wt % according to experimental tests, which melting temperature of the EPCM is 32.17 °C. The application of the prepared EPCMs in textiles can improve the thermal comfort of human body, especially in low temperature environments. Xu et al., [185] prepared binary eutectic phase-change fatty acids composed of LA and SA to block the pores of gold mesoporous silica core-shell nanoparticles, thereby delivering

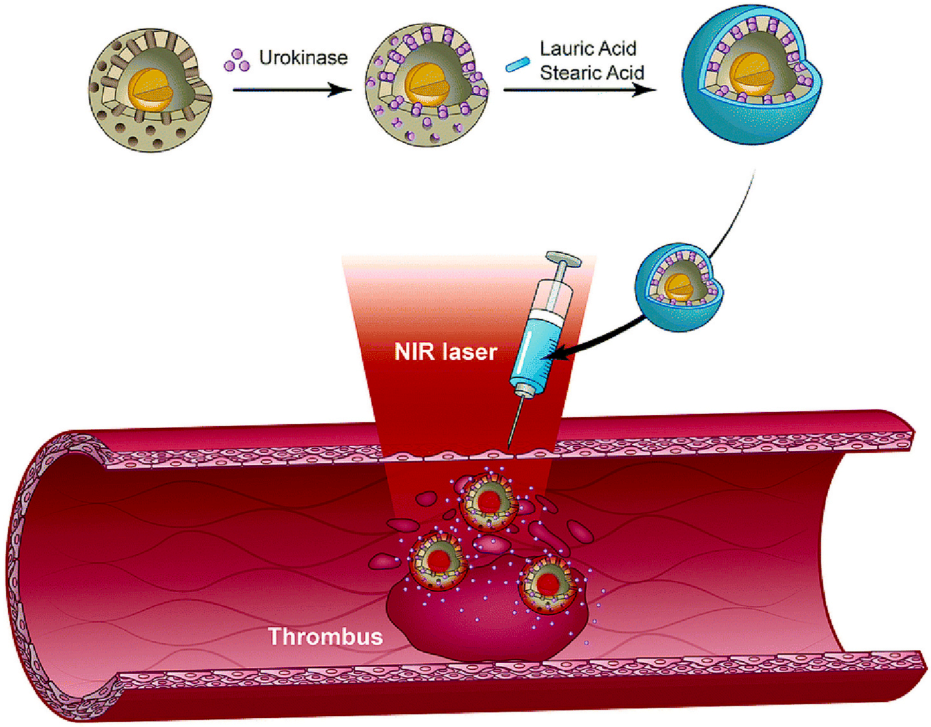

Fig. 12. Application of EPCMs in biological field, adapted from Ref. [185].

thrombolytic drugs. Particles and exhibit a remarkable photothermal effect for fast drug release in response to NIR. In addition, local hyperthermia can also promote the dissolution of thrombus, which can be used for clinical delivery of drugs in the future. In the future, fatty acid EPCMs may be used in the field of clinical drug delivery due to the advantages of wide material sources, non-toxicity, and good biocompatibility. The eutectic nanoparticles are prepared by the method of nanoprecipitation as can be seen in Fig. 12.

5.4. EPCM for PV/T system

Due to the superior temperature stabilizing effect, EPCMs can be used for thermal management of photovoltaic (PV) systems to form PV/T applications. In PV systems, if the temperature of the PV panel is high, the photoelectric conversion efficiency will decrease significantly. When the temperature of PV modules increases by 1 °C, the photoelectric conversion efficiency decreases by 0.3 % ~ 0.5 % [186]. Therefore, it is of great significance to control the temperature and heat dissipation of the PV panel. In order to prevent the increase of temperature from affecting the photoelectric conversion efficiency, Karthikeyan et al., [187] using lauryl alcohol and ethyl alcohol proposed a new type of EPCMs which the phase change temperature is 21.75 °C. The EPCM is used for temperature control of PV modules, the schematic diagram is shown in Fig. 13, a solid EPCM obtained by cooling at night, which absorbs the heat dissipation of PV modules during the daytime. The result of experiment shown that the designed 5 cm thick EPCM container can reduce the module temperature of 10.3 °C and increase the output power by 2.8 %, the average performance is improved by 72.63 %. Compared with organic PCMs, Ramanan et al., [117] prepared inorganic EPCMs which includes $Na_2CO_3{\cdot}10H_2O$ and $MgSO_4{\cdot}7H_2O$. When the mass ratio of the two materials equals to 70 wt%:30 wt%, the phase change temperature of the EPCM is 35.6 °C. Applying these EPCM on the back of the PV photovoltaic module instantly reduces the temperature by 7 °C, the output power increases by 17.63 W, and the daily DC output power increases by 12.5 %.

In the production of hot water from PT system, Mert et al., [188] proposed and screened a eutectic mixture of ammonium aluminum sulfate and sodium hydrogen phosphate in a ratio of 40%wt:60%wt as a binary EPCM with a phase transition temperature of 41.98 °C–60.42 °C, suitable for household solar hot water storage system, using hot water and EPCM as a heat storage tank the experimental structure diagram. The energy and exergy analysis of whether the eutectic material is filled or not shows that the maximum exergy efficiency of the filled EPCM can reach 22 %. Narayanan et al., [16] prepared a 50%wt paraffin-50%wt oleic acid organic EPCM, and 0.5 wt% nano-graphite powder was also added. The EPCM (melting point 53.5 °C, latent heat value 155 kJ/kg) can not only be used for solar hot water heating, but also can be used as a solar heating glove. As see in Fig. 14, during charging progresses, the EPCM stores energy in the form of latent heat (53 ± 0.5 °C) when the composite melts at a constant temperature. During charging, when cold water is circulated through the TES unit, it absorbs thermal energy from the copper pipes through the NGEPCM and produces hot water. In the process of low-temperature solar water production and heat storage, EPCMs with moderate phase transition temperature are selected, such as organic EPCMs [189–191] and inorganic EPCMs [192].

5.5. EPCM for power battery thermal management

EPCMs not only can be used for PV module thermal management, but also play important roles in the thermal management of the power battery of electric vehicles. As a heat storage and heat dissipation material, variable materials have attracted extensive attention of researchers. Ling et al., [193] proposed to prepare inorganic EPCM, mainly based on SAT-Urea, and added EG to EPCM, the experimental results show that for the cooling performance of the battery pack, the inorganic PCM successfully controls the maximum temperature T_{max} of the 20-cell battery pack below 52.3 °C and the maximum temperature difference ΔT_{max} below 4.0 °C. Due to the higher density of latent heat storage, SAT urea performed better in cooling batteries than paraffin with a similar melting point. RyoKoyama et al., [194] measured and studied the thermal properties of trimethylethane (TME) hydrate. When the TME in the prepared TME aqueous solution is 60 wt%, the TME

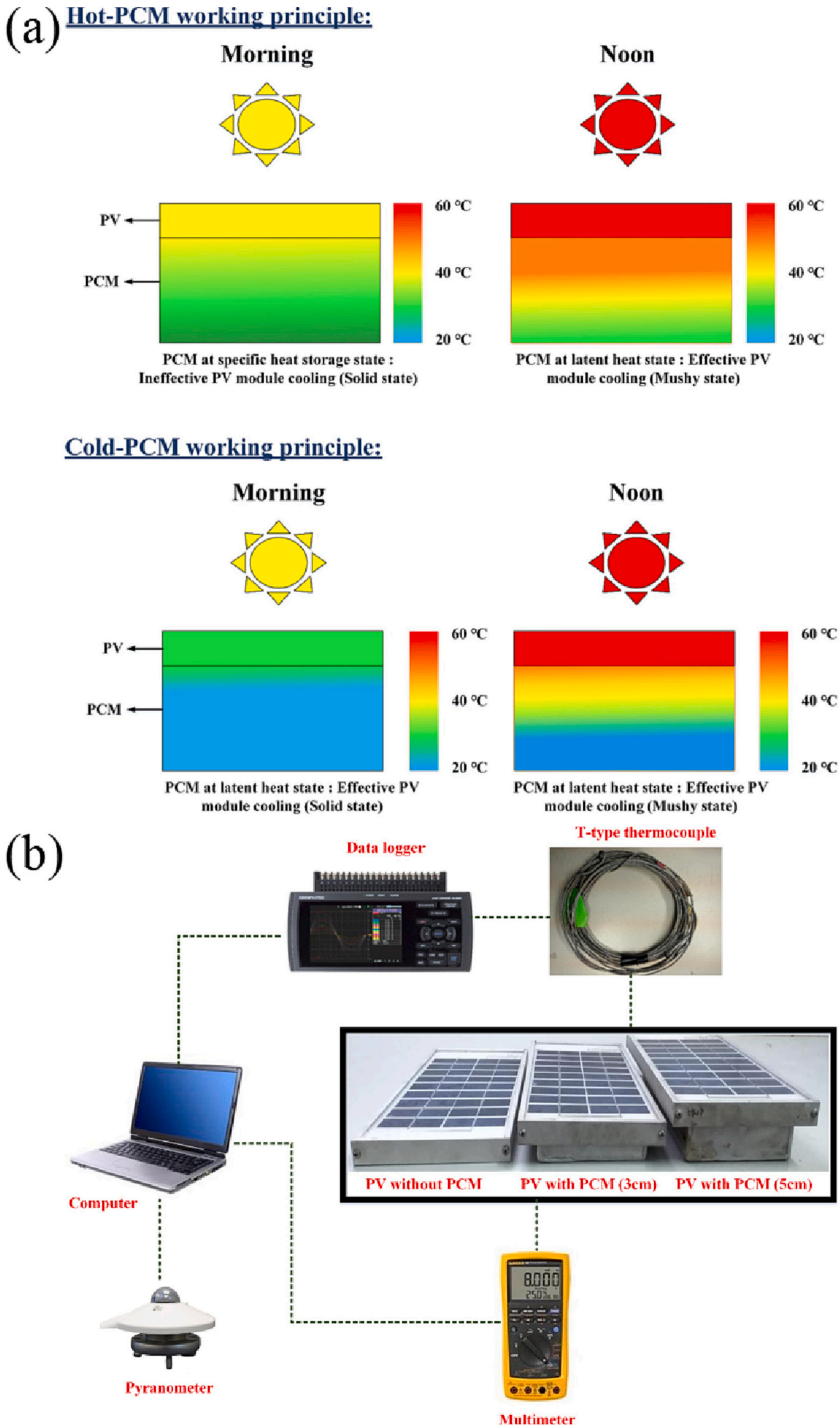

Fig. 13. Preparation of EPCM in PV module, adapted from Ref. [187].

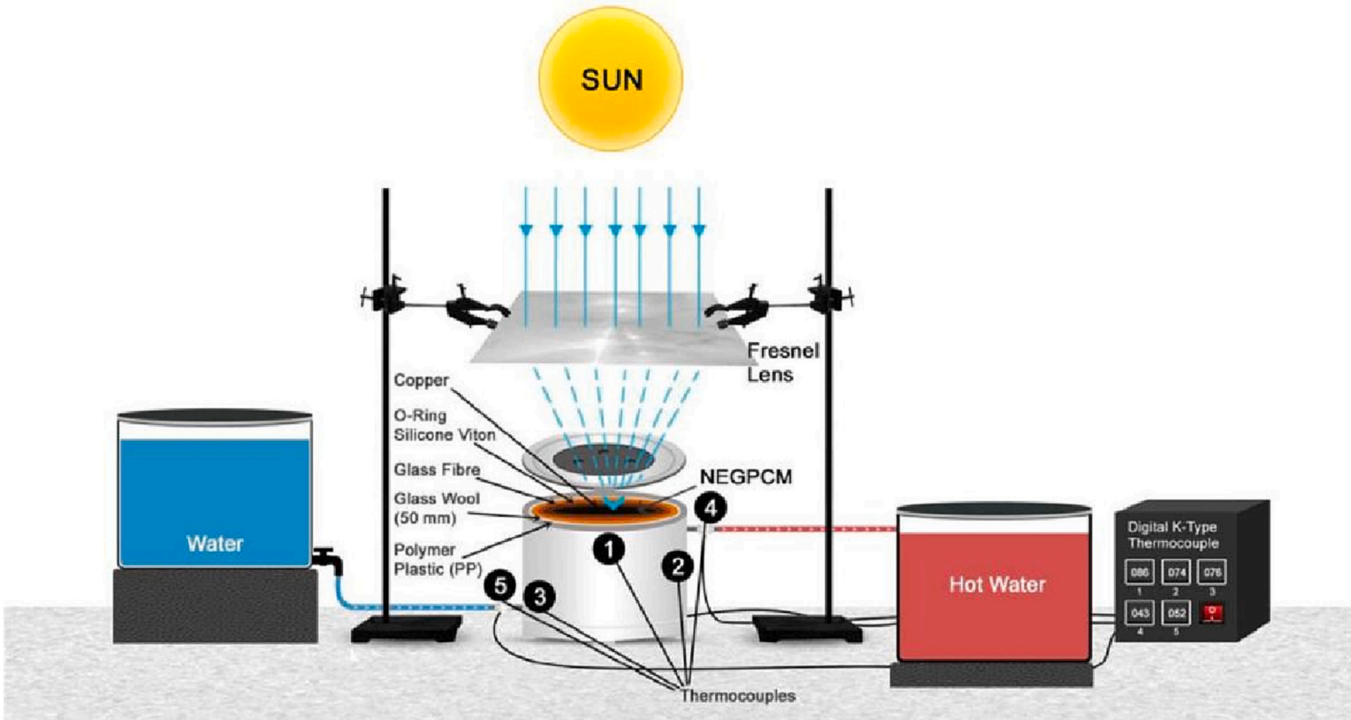

Fig. 14. Schematic diagram of solar water heating system using paraffin EPCMs, adapted from Ref. [16].

hydrate can beobtained by cooling crystallization with a latent heat of 190.1 kJ/kg and a phase transition temperature of 29.6 °C. Therefore, it has a good performance on the cooling of the power battery pack. In terms of battery thermal management in cold environment, Sun et al., [211] proposed to use the subcooling degree of EPCMs to heat the battery pack at a low temperature of −20 °C in winter as shown in Fig. 15a. The ternary brine system prepared by the melt blending method is $Na_2S_2O_3{\cdot}5H_2O{:}CH_3COONa{\cdot}3H_2O{:}H_2O$ with a weight ratio of 64.4 %wt:27.6%wt: 8%wt. The supercooling performance of the EPCM is employed in this study to ensure that the EPCM remains supercooled

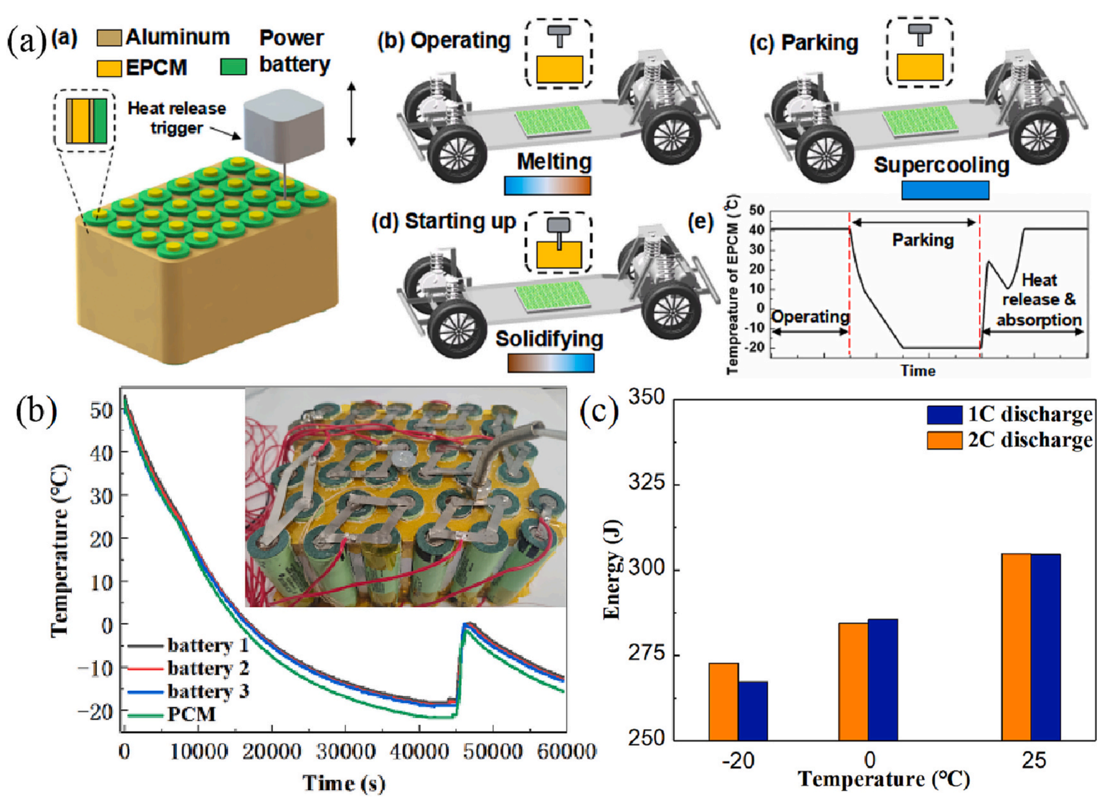

Fig. 15. Preparation of deep supercooling EPCM for BTMS, adapted from Ref. [211].

at −20 °C for long term. The heat can be released by a controllable manner when heat is desired for preheating the power battery in cold environment (Fig. 15b). The discharge efficiency is tested to improve by 6.8 % (Fig. 15c) as compared to the batteries without thermal management.

In the industrial field, EPCMs can not only be used for heat dissipation, but also for heat storage. KHUSHBOO et al., [196] prepared a EPCM by using two organic PCMs, acetanilide:benzoic acid, with a mass ratio of 30 wt%:70 wt%. After a series of thermophysical property tests such as DSC, the phase transition temperature was 75.56 °C, the thermophysical properties are almost stable after 100 cycles of heating and cooling, so the EPCM has good stability when heated and stored in hot water. Tang et al., [197] fabricated shape-stable PCMs using fatty acid eutectic/EG composites and used them for heat storage systems. Fauzi et al., [24] prepared two fatty acid co-crystal mixtures of MA/PA and MA/PA/sodium stearate (SS) respectively. This study shows that MA/PA/SS has better thermophysical properties than MA/PA eutectic mixture after 1500 thermal cycles.

5.6. EPCM for solar-thermal power plants

For the application of EPCMs in high temperature fields, EPCMs could be employed in high temperature power generation and high temperature heat storage stations. There has been a literature [198] that has studied the application of PCMs to high-temperature solar heat storage. By increasing the concentration ratio of materials, especially the mixed nanofluids and molten salts, the absorption of solar energy by storage materials can be improved. The article also discusses high cost, corrosion and erosion, increased pressure drop and coefficient of friction, and instability. In order to avoid the high temperature induced decomposition, corrosion and degradation of the EPCM, alloys or molten salts are typically employed as the EPCMs. In 2014, Blanco Rodríguez et al. [99] proposed an eutectic metal alloy material 49%wt Mg-51%wt Zn with phase transition temperature of 342 °C and latent heat value of 155 kJ/kg, which is used as the EPCM of TES in CSP. Cycling experiments at room temperature and melting temperature were performed, and it was found that this alloy is proposed as a candidate for latent heat energy storage for direct steam generation (DSG) in CSP applications. Risueño et al. [199] aimed at the 70 wt% Mg-24.9 wt% Zn-5.1 wt% Al

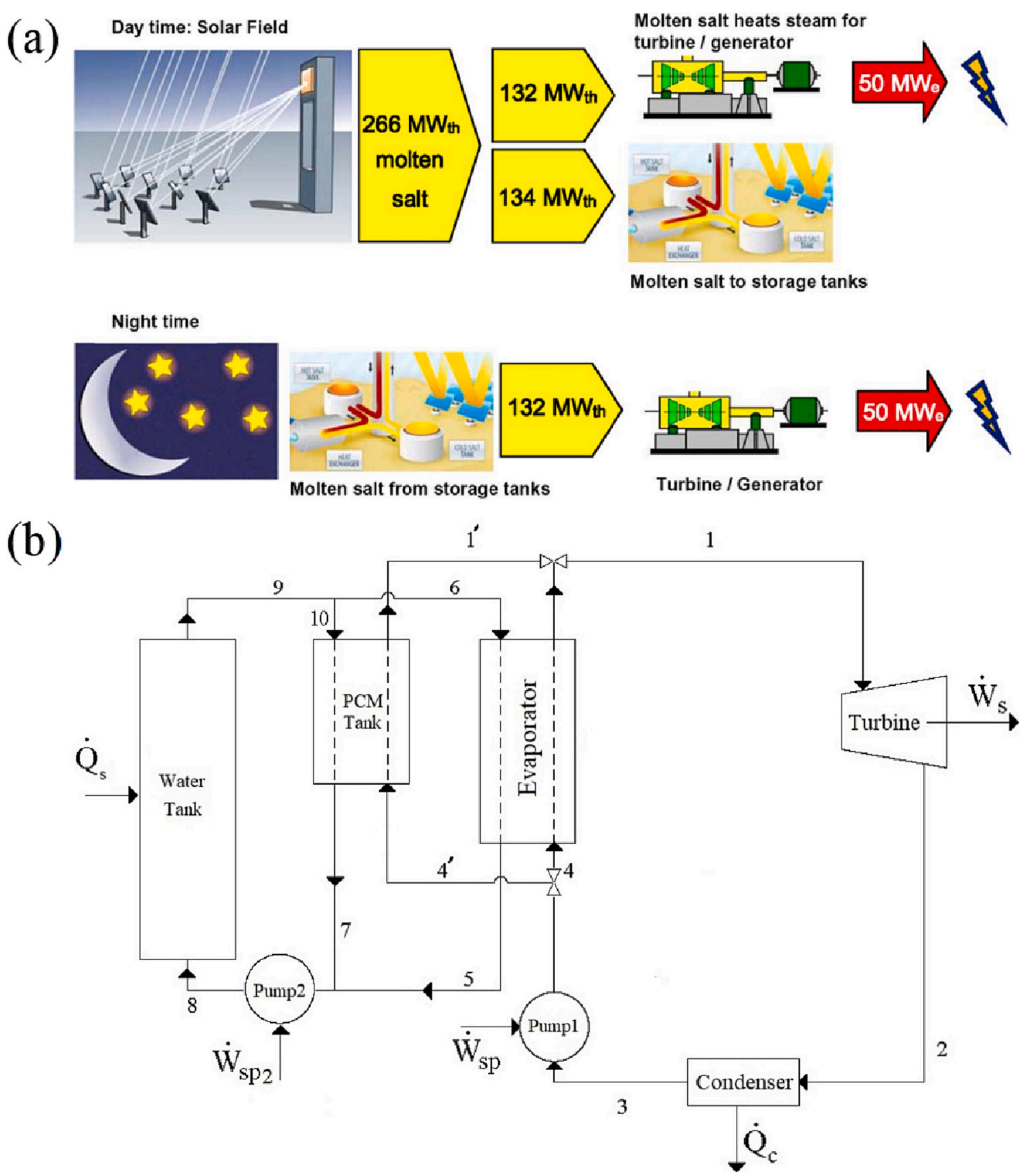

Fig. 16. (a) Solar Salt as EPCM for power generation, adapted from Ref. [206]. (b) Schematic diagram of the modified solar thermal power generation cycle, adapted from Ref. [208].

ternary metal system, the phase transition temperature of the ternary eutectic system is 338.36 °C and has long-term thermal stability. Although metal alloys have high thermal conductivity and have better thermal diffusivity, heat capacity and energy density, compared with molten salts, the biggest disadvantage is high cost and has not been applied on a large scale in industry. Molten salts based EPCMs due to have low cost and high heat storage density, these EPCMs can appropriate applied in CSP system. The common molten salts EPCMs are $NaNO_3$ (60 wt%)-KNO_3 (40 wt%) called *Solar Salt* [200,201], which a commonly used commercial molten-salt in modern CSP systemscan melts at 223 °C, the operation temperature can up to 600 °C stable in liquid. In addition to binary molten salt EPCMs, ternary molten salt EPCMs used in concentrating power generation include: $NaNO_3$ (7 wt %)-KNO_3 (53 wt%)-$NaNO_2$ (40 wt%) called as *Hitec* [202,203] and $NaNO_3$ (7 wt%)-KNO_3 (45 wt%)-$Ca(NO_3)_2$ (48 wt%) called as *Hitec XL* [204,205], the phase transition temperature are also lower than that of the *Solar Salt*, and the maximum temperature does not exceed 500 °C. The use of molten salt EPCMs for CSP has already been commercialized. For example, in early 2011, Solar Salt solar concentrating power generation was completed and put into operation in Spain [206]. *Solar Salt* was used as a EPCM in a CSP system, shown in Fig. 16a, which has a 50 MW_e turbine through a solar concentrating field that a solar receiver will collect at peak solar radiation 266 MW_t of solar heat. Half of the collected heat is fed directly into turbines to generate electricity during the day and producing 50 MW_e of electricity, while the remaining 134 MW_t of thermal energy is stored in molten salt thermal storage tanks. During the night, the thermal energy stored in the molten salt continues to provide heat to the steam turbines, which generate 50 MW_e of electricity. Jiang et al., [207] proposed a low-cost and high melting point (626.0 °C) eutectic Na_2SO_4-NaCl salt to study new high-temperature EPCMs for solar energy storage. The melting point and heat of fusion of the eutectic salt were measured to be 625.3 °C and 263.3 kJ/kg, respectively, and it was found that the addition of α-alumina and mullite to the eutectic Na_2SO_4-NaCl salt could reduce the overheating of the eutectic Na_2SO_4-NaCl salt. Coldness and improved thermal stability have good application prospects for future solar power generation. After the high-temperature molten salt exchanges heat with the heat-conducting oil, the heat-conducting oil exchanges heat with water to generate steam. The introduction of PCMs into building-integrated semi-transparent photovoltaic systems is a promising technology for regulating the surface temperature raised by PV systems [120], $Na_2SO_4·10H_2O$ and $Zn(NO_3)_2·6H_2O$ were mixed at 70 wt%:30 wt% to form binary EPCM by heating and mixing method. The developed eutectic mixture is used in a building-integrated semitransparent photovoltaic - EPCM system to regulate the BISTPV cell temperature. Throughout 2018, the experiment was conducted under outdoor environmental conditions in the region of Kovilpatti, Tamilnadu, India, using this EPCM offered a 12 °C drop in the BISTPV cell temperature which can increase the output power of the BISTPV module from 34,287 W/year to 37,024 W/year.

In conclusion, we have summarized the practical application of EPCMs in various fields in recent years as shown in Table 5. According to the phase transition temperature ranges of the EPCM, previous works have shown that the EPCM could be employed for low, medium and high temperature TES applications. For example, in the aspect of low temperature cold storage, the phase transition temperature of the hydrated salt EPCM is lower than 0 °C, which can be applied to low temperature storage direction. For application in the field of building thermal comfort, it is not only necessary to select a suitable phase transition temperature between 19 °C to 29 °C, but also to select a EPCM with good compatibility with building materials, non-toxic and non-corrosive, such as organic EPCMs. In the biomedical research field, EPCMs are limited by their phase transition temperature and conditions such as non-toxicity, non-corrosiveness, degradability, and harmlessness to the body. Fatty acid-based EPCMs are generally selected with compatible phase transition temperature to human bodies. For the selection of EPCM in the industrial field, in addition to its suitable phase transition temperature, high latent heat value, high thermal conductivity and good cycle stability, properties such as non-corrosive, non-flammable, non-polluting and low-cost should also be taken into consideration.

6. Conclusion and future perspectives

6.1. Future perspective

EPCMs have garnered much attention due to their ability to adjust eutectic temperature and other characteristics by changing the proportion of added components. However, their commercialization is limited by problems such as poor thermal conductivity, supercooling, phase separation, and corrosion. To improve their commercial application, preparing EPCMs with high heat storage density is crucial.

While some PCMs still have a large degree of supercooling, it can be alleviated by preparing EPCMs. Supercooling is a metastable state that is heavily influenced by external conditions. Some studies have explored the use of EPCM's supercooling characteristics for interseasonal energy storage, particularly for inorganic PCM sodium acetate trihydrate and organic PCM erythritol, which have large supercooling characteristics.

To address issues of poor thermal conductivity, phase separation, and corrosion, developing high-performance additives and encapsulating EPCMs are important measures. Although encapsulation may increase costs, it can improve EPCM performance and reduce packaging costs in the future.

Interestingly, the heat storage density of inorganic-organic EPCMs is significantly higher than that of any single-component PCM. This provides a promising direction for preparing phase change materials with high heat storage density in the future.

Given the energy crisis and environmental pollution, it is necessary to research and prepare EPCMs with a wide range of raw materials, high thermal cycle stability, large latent heat value, and low-cost, environmentally friendly features in the future.

7. Conclusion

Through both theoretical and experimental research conducted on binary, ternary, and *n*-nary EPCMs developed in recent years, this review article comprehensively discusses the EPCMs from eutectic theory to the preparation methods. Furthermore, organic-based, inorganic-based, and organic-inorganic EPCMs are carefully reviewed, and the performance enhancement technique for EPCMs and their applications in diverse fields are comprehensively presented. Based on the findings of the research, the following conclusions can be drawn:

- Different organic or inorganic PCMs can be selected for specific applications according to eutectic theory and phase diagram analysis. The phase change temperature range is extended, and the latent heat value of the EPCMs can be varied as compared to single-component PCM.
- A review of the prepared EPCMs indicates that >250 EPCMs have been reported so far, with phase transition temperature ranges from −15 °C to 1157 °C and latent heat values ranging from 90.2 kJ/kg to 640 kJ/kg. For organic-organic EPCMs, due to the low phase transition temperature of single-component organic PCM, the phase transition temperature of the organic EPCMs range from 6.5 °C to 178.3 °C, and the latent heat value is 90.2 kJ/kg to 332.3 kJ/kg. However, the inorganic PCM, molten salt, and metal alloy have high phase transition temperature, resulting in high phase transition temperature of molten salt and metal alloy-based EPCM and a large phase transition latent heat value. For the hydrated salt PCM, the phase transition temperature of the brine system EPCMs composed of its aqueous solution is lower, which can reach −15 °C.
- Organic EPCMs suffer from poor thermal conductivity and easy leakage after phase transition. Carbon-based and metal-based

Table 5
Practical Applications of EPCMs in various fields.

Application number	EPCMs composition	Eutectic phase change temperature (°C)	Latent heat (kJ/kg)	Applications	Reference
EPCMs using in cold storage					
1	30 wt% PEG200-70 wt% PEG300	−20	–	EPCM used in the refrigerator can reduce the temperature fluctuation by 40.59 % with 2.0 kg EPCM and the energy consumption of the refrigerator EPCM can save the 8.37 % lower than ordinary refrigerator.	[172]
2	66 wt% Tetradecane-34 wt% Lauryl alcohol	4.3	247.1	The average cold keeping time in the empty vaccine refrigerator adding EPCM at 2 °C ~ 8 °C was 44.37 h, and the average holding time during the experiment was 46.04 h when the load was added.	[175]
3	19.5 % NH_4Cl solution	−15.4	–	Adding EPCM to the freezer reduced energy consumption by 8 % during the defrost cycle and by 7 % during the system's door opening period.	[173]
4	84 wt% Decyl alcohol-16 wt% Lauric acid	2.08	188.71	When the ambient temperature is 3 °C, the cold storage time of the incubator filled with EPCM can be close to 10 h.	[176]
5	19.5 wt% NH4Cl solution	−15.6	–	The compressor start-stop time was reduced by 9 % with the addition of EPCMs and the energy consumption of the evaporator was reduced by 5.81 %.	[174]
EPCMs using in building					
6	40 wt% LA-60 wt% Myristyl alcohol	21.3	151.5	After 1000 cycles, the EPCM was compatible with Al and SS316, but shows corrosion with Cu which is a good candidate for improving building thermal comfort.	[51]
7	70 wt% CA 70 %-30 wt% cetyl alcohol	22.89	144.92	After 1000 cycles the EPCM was found without significant change and EPCM also have good compatibility with copper, aluminum and stainless steel 316	[179]
8	94 wt% $CaCl_2 \cdot 6H_2O$-6 wt% $Mg(NO_3)_2 \cdot 6H_2O$	27.51	142.4	Compared with the reference room, when the EPCM thickness is 30 mm, the energy consumption can be saved by 190.05 kWh in summer and 361.46 kWh in winter	[180]
9	73.4 wt% LA-26.6 wt% PA	36	187	The use of nano-silica-loaded EPCMs in building materials can store a large amount of thermal energy and maintain thermal comfort to the greatest extent	[152]
10	55 wt% Erythritol-45 wt% urea	68.6	220.54	The EPCM integrated solar heating system can save 5 tons of standard coal in one heating season for a building with a heating area of 1125 m^2 in Tianjin, China.	[181]
EPCMs using in Biomedical					
11	80 wt% LA-20 wt% SA	39	–	After the dye absorbs near-infrared light, according to the photothermal effect, the temperature rises to melt the EPCM, thereby releasing the drug.	[183]
12	69 wt% LA-31 wt% PA	32.17	166.9	A new material for energy management, EPCM was developed by coaxial electrospinning which can improve the thermal comfort of human body, especially in low temperature environments.	[209]
EPCMS using in PV/T system					
13	75 wt% Lauryl alcohol-25 wt% Ethyl alcohol	21.75	199	Designed 5 cm thick EPCM container can reduce the module temperature of 10.3 °C and increase the output power by 2.8 %, the average performance is improved by 72.63 %.	[187]
14	70 wt% $Na_2CO_3 \cdot 10H_2O$-30 wt% $MgSO_4 \cdot 7H_2O$	35.6	230.5	Applying EPCM on the back of the PV module instantly reduces the temperature by 7 °C, the output power increases by 17.63 W and the daily DC output power increases by 12.5 %.	[117]
15	40 wt% ammonium aluminum sulfate-60 wt% sodium hydrogen phosphate	41.98–60.42	288.13	Energy and exergy analysis shown that the heat storage tank with the EPCM is more efficient than without the EPCM and the maximum exergy efficiency was obtained as 22 %.	[188]
16	50 wt% paraffin-50 wt% oleic acid	53.5	155	Not only be used for solar hot water heating, but also can be used as a solar heating glove.	[16]
EPCMs using for BTMs					
17	91 wt% SAT-9 wt% Urea	50.3	181	For the cooling performance of the battery pack, these EPCM successfully controls the maximum temperature T_{max} of the battery pack below 52.3 °C and the maximum temperature difference ΔT_{max} below 4.0 °C.	[193]
18	60 wt% Trimethylethane (TME) hydrate	29.6	190.1	When used as a cooling medium for lithium-ion batteries, EPCM has more than three times the energy density of liquid water and is corrosion-resistant, non-flammable.	[210]
19	64.6 wt% $Na_2S_2O_3 \cdot 5H_2O$-27.6 wt% $CH_3COONa \cdot 3H_2O$-8 wt% H_2O	41.4	169.29	Utilizing the large supercooling properties of EPCM and the discharge efficiency is tested to improve by 6.8 % as compared to the batteries without thermal management.	[211]
EPCMs using in solar power generation					
20	49 wt% Mg-51 wt% Zn	342	155	Alloy is proposed as a candidate for latent heat energy storage for DSG in CSP applications.	[99]

(continued on next page)

Table 5 (*continued*)

Application number	EPCMs composition	Eutectic phase change temperature (°C)	Latent heat (kJ/kg)	Applications	Reference
21	60 wt% $NaNO_3$-40 wt% KNO_3	223	–	Half of the collected heat is fed directly into turbines to generate electricity during the day and producing 50 MW_e of electricity. During the night, the thermal energy continues to provide heat to the steam turbines, which generate 50 MW_e of electricity.	[206]
22	68.05 wt% Na_2SO_4–31.95 wt% NaCl	626.0	263.3	After the high-temperature molten salt exchanges heat with the heat-conducting oil, the oil exchanges heat with water to generate steam.	[207]
23	70 wt% $Na_2SO_4 \cdot 10H_2O$-30 wt% $Zn(NO_3)_2 \cdot 6H_2O$	30.0	242	Using these EPCM can increase the output power of the BISTPV module from 34,287 W/year to 37,024 W/year.	[120]

additives are commonly employed to improve the thermal conductivity of organic EPCMs. Inorganic EPCMs have obvious supercooling and phase separation phenomena in brine systems. The commonly used method is to add nucleating agents and thickeners to improve supercooling and inhibit phase separation.
- By improving the performance of EPCMs, it can be widely used in various application fields such as low-temperature cold storage, building thermal comfort, biomedical sciences, PV/T system, power battery thermal management as well as solar-thermal power plants. There is a need to develop more types of EPCMs and new applications in the future.

CRediT authorship contribution statement

Mingyang Sun: drafted the manuscript, Writing – review & editing. Tong Liu, Mulin Li, Tianze Liu and Xinlei Wang: helped to collect the literatures. Guijun Chen and Jiadian Wang: Writing – review & editing. Dongyue Jiang: Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by National Natural Science Foundation of China (51906031) Fundamental Research Funds for the Central Universities, DUT22YG108.

References

[1] H. Nazir, M. Batool, F.J. Bolivar Osorio, M. Isaza-Ruiz, X. Xu, K. Vignarooban, et al., Recent developments in phase change materials for energy storage applications: a review, Int. J. Heat Mass Transf. 129 (2019) 491–523.
[2] X.-Z. Lan, Z.-C. Tan, D.-T. Yue, Q. Shi, C.-G. Yang, Heat storage performance of disodium hydrogen phosphate dodecahydrate: prevention of phase separation by thickening and gelling methods, Chin. J. Chem. 25 (2007) 921–925.
[3] N. Kumar, D. Banerjee, A comprehensive review of salt hydrates as phase change materials (PCMs), Int. J. Transp. Phenom. 15 (2018) 65–89.
[4] N. Radouane, A comprehensive review of composite phase change materials (cPCMs) for thermal management applications, including manufacturing processes, performance, and applications, Energies 15 (2022) 8271.
[5] C. Zhu, D. Huo, Q. Chen, J. Xue, S. Shen, Y. Xia, A eutectic mixture of natural fatty acids can serve as the gating material for near-infrared-triggered drug release, Adv. Mater. 29 (2017) 1703702.
[6] Y. Liu, Y. Yang, Preparation and thermal properties of Na2CO3·10H2O-Na2HPO4·12H2O eutectic hydrate salt as a novel phase change material for energy storage, Appl. Therm. Eng. 112 (2017) 606–609.
[7] Y. Liu, Y. Yang, Form-stable phase change material based on Na2CO3·10H2O-Na2HPO4·12H2O eutectic hydrated salt/expanded graphite oxide composite: the influence of chemical structures of expanded graphite oxide, Renew. Energy 115 (2018) 734–740.
[8] Y. Yuan, N. Zhang, W. Tao, X. Cao, Y. He, Fatty acids as phase change materials: a review, Renew. Sust. Energ. Rev. 29 (2014) 482–498.
[9] B. Cárdenas, N. León, High temperature latent heat thermal energy storage: phase change materials, design considerations and performance enhancement techniques, Renew. Sust. Energ. Rev. 27 (2013) 724–737.
[10] P. Singh, R.K. Sharma, A.K. Ansu, R. Goyal, A. Sarı, V.V. Tyagi, A comprehensive review on development of eutectic organic phase change materials and their composites for low and medium range thermal energy storage applications, Sol. Energy Mater. Sol. Cells 223 (2021), 110955.
[11] A.A.A. Abuelnuor, A.A.M. Omara, K.M. Saqr, I.H.I. Elhag, Improving indoor thermal comfort by using phase change materials: a review, Int. J. Energy Res. 42 (2018) 2084–2103.
[12] R.A. Lawag, H.M. Ali, Phase change materials for thermal management and energy storage: a review, J. Energy Storage 55 (2022), 105602.
[13] J. Huang, Y. Luo, M. Weng, J. Yu, L. Sun, H. Zeng, et al., Advances and applications of phase change materials (PCMs) and PCMs-based technologies , ES Mater. Manuf. 13 (2021) 23–39.
[14] A. Sarı, C. Alkan, C. Bilgin, Micro/nano encapsulation of some paraffin eutectic mixtures with poly(methyl methacrylate) shell: preparation, characterization and latent heat thermal energy storage properties, Appl. Energy 136 (2014) 217–227.
[15] L. Li, G. Wang, C. Guo, Influence of intumescent flame retardant on thermal and flame retardancy of eutectic mixed paraffin/polypropylene form-stable phase change materials, Appl. Energy 162 (2016) 428–434.
[16] S.S. Narayanan, A. Kardam, V. Kumar, N. Bhardwaj, D. Madhwal, P. Shukla, et al., Development of sunlight-driven eutectic phase change material nanocomposite for applications in solar water heating, Resource-Efficient Technol. 3 (2017) 272–279.
[17] A.C. Rustan, C.A. Drevon, Fatty Acids: Structures and Properties, eLS, 2005.
[18] D. Rozanna, T.G. Chuah, A. Salmiah, T.S.Y. Choong, M. Sa'ari, Fatty acids as phase change materials (PCMs) for thermal energy storage: a review, Int. J. Green Energy 1 (2005) 495–513.
[19] H. Ke, Phase diagrams, eutectic mass ratios and thermal energy storage properties of multiple fatty acid eutectics as novel solid-liquid phase change materials for storage and retrieval of thermal energy, Appl. Therm. Eng. 113 (2017) 1319–1331.
[20] L. Xiao, R. Ding, H. Sun, F. Wang, A new kind of shape-stabilized phase change materials, J. Wuhan Univ. Technol. Mater. Sci. Ed. 26 (2011) 491–494.
[21] H. Nazir, M. Batool, M. Ali, A.M. Kannan, Fatty acids based eutectic phase change system for thermal energy storage applications, Appl. Therm. Eng. 142 (2018) 466–475.
[22] A. Sarı, Eutectic mixtures of some fatty acids for low temperature solar heating applications: thermal properties and thermal reliability, Appl. Therm. Eng. 25 (2005) 2100–2107.
[23] H. He, Q. Yue, B. Gao, X. Zhang, Q. Li, Y. Wang, The effects of compounding conditions on the properties of fatty acids eutectic mixtures as phase change materials, Energy Convers. Manag. 69 (2013) 116–121.
[24] H. Fauzi, H.S.C. Metselaar, T.M.I. Mahlia, M. Silakhori, Sodium laurate enhancements the thermal properties and thermal conductivity of eutectic fatty acid as phase change material (PCM), Sol. Energy 102 (2014) 333–337.
[25] L. Yao, B. Yang, H. Cui, J. Zhuang, J. Ye, J. Xue, Challenges and progresses of energy storage technology and its application in power systems, J. Mod. Power Syst. Clean Energy 4 (2016) 519–528.
[26] S. Tomassetti, A. Aquilanti, P.F. Muciaccia, G. Coccia, C. Mankel, E.A. B. Koenders, et al., A review on thermophysical properties and thermal stability of sugar alcohols as phase change materials, J. Energy Storage 55 (2022), 105456.
[27] G. Diarce, A. Rojo, L. Quant, L. Bouzas, A. García-Romero, Thermal endurance of xylitol as a phase change material for thermal energy storage applications, J. Energy Storage 55 (2022), 105717.
[28] X.-F. Shao, C. Wang, Y.-J. Yang, B. Feng, Z.-Q. Zhu, W.-J. Wang, et al., Screening of sugar alcohols and their binary eutectic mixtures as phase change materials for low-to-medium temperature latent heat storage. (I): non-isothermal melting and crystallization behaviors, Energy 160 (2018) 1078–1090.
[29] J.-L. Zeng, Y.-H. Chen, L. Shu, L.-P. Yu, L. Zhu, L.-B. Song, et al., Preparation and thermal properties of exfoliated graphite/erythritol/mannitol eutectic composite as form-stable phase change material for thermal energy storage, Sol. Energy Mater. Sol. Cells 178 (2018) 84–90.
[30] J. Zuo, W. Li, L. Weng, Thermal performance of caprylic acid/1-dodecanol eutectic mixture as phase change material (PCM), Energy Build. 43 (2011) 207–210.

[31] Y. Cai, G. Sun, M. Liu, J. Zhang, Q. Wang, Q. Wei, Fabrication and characterization of capric–lauric–palmitic acid/electrospun SiO2 nanofibers composite as form-stable phase change material for thermal energy storage/retrieval, Sol. Energy 118 (2015) 87–95.
[32] R. Kumar, S. Vyas, A. Dixit, Fatty acids/1-dodecanol binary eutectic phase change materials for low temperature solar thermal applications: design, development and thermal analysis, Sol. Energy 155 (2017) 1373–1379.
[33] E.B. Jebasingh, Effects of various types of graphite on the thermal conductivity and energy storage properties of ternary eutectic fatty acid-based composite as phase change material, Renewables Wind Water Solar 3 (2016) 8.
[34] S. Imran Hussain, A. Ameelia Roseline, S. Kalaiselvam, Enhancement of thermal conductivity and thermal stability of capric-lauric acid eutectic phase change material using carbonaceous materials, Mater. Res. Express 6 (2019), 115605.
[35] L. Shilei, Z. Neng, F. Guohui, Eutectic mixtures of capric acid and lauric acid applied in building wallboards for heat energy storage, Energy Build. 38 (2006) 708–711.
[36] L. Wang, D. Meng, Fatty acid eutectic/polymethyl methacrylate composite as form-stable phase change material for thermal energy storage, Appl. Energy 87 (2010) 2660–2665.
[37] H. Wei, X. Xie, X. Li, X. Lin, Preparation and characterization of capric-myristic-stearic acid eutectic mixture/modified expanded vermiculite composite as a form-stable phase change material, Appl. Energy 178 (2016) 616–623.
[38] J. Huang, S. Lu, X. Kong, S. Liu, Y. Li, Form-stable phase change materials based on eutectic mixture of tetradecanol and fatty acids for building energy storage: preparation and performance analysis, Materials (Basel) 6 (2013) 4758–4775.
[39] Z. Luo, H. Zhang, X. Gao, T. Xu, Y. Fang, Z. Zhang, Fabrication and characterization of form-stable capric-palmitic-stearic acid ternary eutectic mixture/nano-SiO2 composite phase change material, Energy Build. 147 (2017) 41–46.
[40] H. Zhang, X. Gao, C. Chen, T. Xu, Y. Fang, Z. Zhang, A capric–palmitic–stearic acid ternary eutectic mixture/expanded graphite composite phase change material for thermal energy storage, Compos. A: Appl. Sci. Manuf. 87 (2016) 138–145.
[41] D. Meng, K. Zhao, W. Zhao, G. Jiang, Preparation and characterization of CA-MA eutectic/silicon dioxide nanoscale composite phase change material from water glass via sol-gel method, J. Wuhan Univ. Technol. Mater. Sci. Ed. 32 (2017) 1048–1056.
[42] D. Zhou, Y. Zhou, Y. Liu, X. Luo, J. Yuan, J. Siddiqui, Preparation and performance of capric-myristic acid binary eutectic mixtures for latent heat thermal energy storages, J. Nanomater. 2019 (2019) 9.
[43] D. Meng, L. Wang, Characterization and thermal conductivity of modified graphite/fatty acid eutectic/PMMA form-stable phase change material, J. Wuhan Univ. Technol. Mater. Sci. Ed. 28 (2013) 586–591.
[44] A. Karaipekli, A. Sarı, Capric–myristic acid/expanded perlite composite as form-stable phase change material for latent heat thermal energy storage, Renew. Energy 33 (2008) 2599–2605.
[45] A. Karaipekli, A. Sarı, Capric–myristic acid/vermiculite composite as form-stable phase change material for thermal energy storage, Sol. Energy 83 (2009) 323–332.
[46] A. Sarı, A. Karaipekli, Preparation and thermal properties of capric acid/palmitic acid eutectic mixture as a phase change energy storage material, Mater. Lett. 62 (2008) 903–906.
[47] F. Tang, D. Su, Y. Tang, G. Fang, Synthesis and thermal properties of fatty acid eutectics and diatomite composites as shape-stabilized phase change materials with enhanced thermal conductivity, Sol. Energy Mater. Sol. Cells 141 (2015) 218–224.
[48] C. Veerakumar, A. Sreekumar, Preparation, thermophysical studies, and corrosion analysis of a stable capric acid/cetyl alcohol binary eutectic phase change material for cold thermal energy storage, Energy Technol. 6 (2018) 397–405.
[49] S. Kahwaji, M.A. White, Prediction of the properties of eutectic fatty acid phase change materials, Thermochim. Acta 660 (2018) 94–100.
[50] S. Kahwaji, M.B. Johnson, A.C. Kheirabadi, D. Groulx, M.A. White, Stable, low-cost phase change material for building applications: the eutectic mixture of decanoic acid and tetradecanoic acid, Appl. Energy 168 (2016) 457–464.
[51] V. Chinnasamy, S. Appukuttan, Preparation and thermal properties of lauric acid/myristyl alcohol as a novel binary eutectic phase change material for indoor thermal comfort, Energy Storage 1 (2019), e80.
[52] R.M. Saeed, J.P. Schlegel, C. Castano, R. Sawafta, V. Kuturu, Preparation and thermal performance of methyl palmitate and lauric acid eutectic mixture as phase change material (PCM), J. Energy Storage 13 (2017) 418–424.
[53] N. Zhang, Y. Yuan, X. Wang, X. Cao, X. Yang, S. Hu, Preparation and characterization of lauric–myristic–palmitic acid ternary eutectic mixtures/expanded graphite composite phase change material for thermal energy storage, Chem. Eng. J. 231 (2013) 214–219.
[54] L. Cao, Y. Tang, G. Fang, Preparation and properties of shape-stabilized phase change materials based on fatty acid eutectics and cellulose composites for thermal energy storage, Energy. 80 (2015) 98–103.
[55] B. Hemmatian, N. Heidarzadeh, G.C. Fard, L. Maleknia, Fabrication of phase-change core/shell nanofibers based on a eutectic fatty acid mixture to control body temperature fluctuations, Mater. Chem. Phys. 245 (2020), 122738.
[56] X. Zhang, Z. Huang, Z. Yin, W. Zhang, Y. Huang, Y. Liu, et al., Form stable composite phase change materials from palmitic-lauric acid eutectic mixture and carbonized abandoned rice: preparation, characterization, and thermal conductivity enhancement, Energy Build. 154 (2017) 46–54.
[57] S. Keleş, K. Kaygusuz, A. Sarı, Lauric and myristic acids eutectic mixture as phase change material for low-temperature heating applications, Int. J. Energy Res. 29 (2005) 857–870.
[58] L. Ma, C. Guo, R. Ou, L. Sun, Q. Wang, L. Li, Preparation and characterization of modified porous wood flour/lauric-myristic acid eutectic mixture as a form-stable phase change material, Energy Fuel 32 (2018) 5453–5461.
[59] L. Han, G. Ma, S. Xie, J. Sun, Y. Jia, Y. Jing, Thermal properties and stabilities of the eutectic mixture: 1,6-hexanediol/lauric acid as a phase change material for thermal energy storage, Appl. Therm. Eng. 116 (2017) 153–159.
[60] W. Zhang, X. Zhang, X. Zhang, Z. Yin, Y. Liu, M. Fang, et al., Lauric-stearic acid eutectic mixture/carbonized biomass waste corn cob composite phase change materials: preparation and thermal characterization, Thermochim. Acta 674 (2019) 21–27.
[61] A. Sari, K. Kaygusuz, Thermal performance of a eutectic mixture of lauric and stearic acids as PCM encapsulated in the annulus of two concentric pipes, Sol. Energy 72 (2002) 493–504.
[62] H. Yang, W.-h. Chen, X.-f. Kong, X. Rong, Fabrication, property characterization and thermal performance of composite phase change material plates based on tetradecanol-myristic acid binary eutectic mixture/expanded perlite and expanded vermiculite for building application, J. Cent. South Univ. 26 (2019) 2578–2595.
[63] A. Sarı, Thermal characteristics of a eutectic mixture of myristic and palmitic acids as phase change material for heating applications, Appl. Therm. Eng. 23 (2003) 1005–1017.
[64] G. Alva, X. Huang, L. Liu, G. Fang, Synthesis and characterization of microencapsulated myristic acid–palmitic acid eutectic mixture as phase change material for thermal energy storage, Appl. Energy 203 (2017) 677–685.
[65] H. Fauzi, H.S.C. Metselaar, T.M.I. Mahlia, M. Silakhori, H. Nur, Phase change material: optimizing the thermal properties and thermal conductivity of myristic acid/palmitic acid eutectic mixture with acid-based surfactants, Appl. Therm. Eng. 60 (2013) 261–265.
[66] I.H. S, A.R. A, K. S, Bifunctional nanoencapsulated eutectic phase change material core with SiO2/SnO2 nanosphere shell for thermal and electrical energy storage, Mater. Des. 154 (2018) 291–301.
[67] Z. Hu, C. Wang, W. Jia, X. Li, Z. Cai, Preparation and thermal properties of 1-hexadecanol-palmitic acid eutectic mixture/activated carbon composite phase change material for thermal energy storage, ChemistrySelect 4 (2019) 222–227.
[68] G. Baran, A. Sari, Phase change and heat transfer characteristics of a eutectic mixture of palmitic and stearic acids as PCM in a latent heat storage system, Energy Convers. Manag. 44 (2003) 3227–3246.
[69] P. Gallart-Sirvent, M. Martín, G. Villorbina, M. Balcells, A. Solé, C. Barrenche, et al., Fatty acid eutectic mixtures and derivatives from non-edible animal fat as phase change materials, RSC Adv. 7 (2017) 24133–24139.
[70] N. Zhang, Y. Yuan, Y. Yuan, X. Cao, X. Yang, Effect of carbon nanotubes on the thermal behavior of palmitic–stearic acid eutectic mixtures as phase change materials for energy storage, Sol. Energy 110 (2014) 64–70.
[71] N. Zhang, Y. Yuan, Y. Du, X. Cao, Y. Yuan, Preparation and properties of palmitic-stearic acid eutectic mixture/expanded graphite composite as phase change material for energy storage, Energy 78 (2014) 950–956.
[72] J. Shen, Z. Cai, C. Wang, X. Liu, R. Zheng, Preparation and thermal performances of 1, 10-decanediol-stearic acid eutectic as phase change material, Thermochim. Acta 690 (2020), 178648.
[73] G. Ma, J. Sun, Y. Zhang, Y. Jing, Y. Jia, A novel low-temperature phase change material based on stearic acid and hexanamide eutectic mixture for thermal energy storage, Chem. Phys. Lett. 714 (2019) 166–171.
[74] G. Ma, L. Han, J. Sun, Y. Jia, Thermal properties and reliability of eutectic mixture of stearic acid-acetamide as phase change material for latent heat storage, J. Chem. Thermodyn. 106 (2017) 178–186.
[75] G. Ma, S. Liu, S. Xie, Y. Jing, Q. Zhang, J. Sun, et al., Binary eutectic mixtures of stearic acid-n-butyramide/n-octanamide as phase change materials for low temperature solar heat storage, Appl. Therm. Eng. 111 (2017) 1052–1059.
[76] G. Ma, J. Sun, Y. Zhang, Y. Jing, Y. Jia, Preparation and thermal properties of stearic acid-benzamide eutectic mixture/expanded graphite composites as phase change materials for thermal energy storage, Powder Technol. 342 (2019) 131–140.
[77] G. Ma, J. Sun, S. Xie, Z. Wang, Y. Jing, Y. Jia, Solid-liquid phase equilibria of stearic acid and dicarboxylic acids binary mixtures as low temperature thermal energy storage materials, J. Chem. Thermodyn. 120 (2018) 60–71.
[78] G. Ma, Z. Wang, S. Xie, J. Sun, Y. Jia, Y. Jing, et al., Preparation and properties of stearic acid–acetanilide eutectic mixture/expanded graphite composite phase-change material for thermal energy storage, Energy Technol. 6 (2018) 153–160.
[79] K. Purohit, V.V.S. Murty, R.C. Dixit, A. Sharma, Development of an acetanilide/benzoic acid eutectic phase change material based thermal energy storage unit for a passive water heating system, Bull. Mater. Sci. 42 (2019) 119.
[80] S. Xie, J. Sun, Z. Wang, S. Liu, L. Han, G. Ma, et al., A thermally stable phase change material with high latent heat based on an oxalic acid dihydrate/boric acid binary eutectic system, Sol. Energy Mater. Sol. Cells 168 (2017) 38–44.
[81] Y. Seki, Ş. İnce, M.A. Ezan, A. Turgut, A. Erek, Graphite nanoplates loading into eutectic mixture of Adipic acid and Sebacic acid as phase change material, Sol. Energy Mater. Sol. Cells 140 (2015) 457–463.
[82] S. Liu, L. Han, S. Xie, Y. Jia, J. Sun, Y. Jing, et al., A novel medium-temperature form-stable phase change material based on dicarboxylic acid eutectic mixture/expanded graphite composites, Sol. Energy 143 (2017) 22–30.
[83] A. Paul, L. Shi, C.W. Bielawski, A eutectic mixture of galactitol and mannitol as a phase change material for latent heat storage, Energy Convers. Manag. 103 (2015) 139–146.

[84] M.M. Kenisarin, High-temperature phase change materials for thermal energy storage, Renew. Sust. Energ. Rev. 14 (2010) 955–970.
[85] K.T. Fujimoto, L.A. Hone, K.D. Manning, R.D. Seifert, K.L. Davis, J.N. Milloway, et al., Additive manufacturing of miniaturized peak temperature monitors for in-pile applications, in: Sensors, 2021.
[86] C.C. Yang, G. Li, Q. Jiang, Effect of pressure on melting temperature of silicon, J. Phys. Condens. Matter 15 (2003) 4961.
[87] E. Risueño, A. Faik, J. Rodríguez-Aseguinolaza, P. Blanco-Rodríguez, A. Gil, M. Tello, et al., Mg-Zn-Al eutectic alloys as phase change material for latent heat thermal energy storage, Energy Procedia 69 (2015) 1006–1013.
[88] E. Risueño, A. Gil, J. Rodríguez-Aseguinolaza, A. Gil, M. Tello, A. Faik, et al., Thermal cycling testing of Zn–Mg–Al eutectic metal alloys as potential high-temperature phase change materials for latent heat storage, J. Therm. Anal. Calorim. 129 (2017) 885–894.
[89] D. Farkas, C.E. Birchenall, New eutectic alloys and their heats of transformation, Metall. Trans. A. 16 (1985) 323–328.
[90] T. Akiyama, Y. Ashizawa, J.-i. Yagi, Storage and release of heat in a single spherical capsule containing phase-change material with a high melting point, Heat Transf. Jap. Res. 21 (1991) 199–217.
[91] C. Chiappe, M. Malvaldi, B. Melai, S. Fantini, U. Bardi, S. Caporali, An unusual common ion effect promotes dissolution of metal salts in room-temperature ionic liquids: a strategy to obtain ionic liquids having organic–inorganic mixed cations, Green Chem. 12 (2010) 77–80.
[92] L. Gang, B. Zhang, L. Xiang, Z. Yuan, Y. Qiang, The preparation, characterization and modification of a new phase change material: CaCl2·6H2O–MgCl2·6H2O eutectic hydrate salt, Sol. Energy Mater. Sol. Cells 126 (2014) 51–55.
[93] Y. Liu, Y. Yang, Use of nano-α-Al2O3 to improve binary eutectic hydrated salt as phase change material, Sol. Energy Mater. Sol. Cells 160 (2017) 18–25.
[94] H. Schmit, C. Rathgeber, P. Hennemann, S. Hiebler, Three–step method to determine the eutectic composition of binary and ternary mixtures, J. Therm. Anal. Calorim. 117 (2014) 595–602.
[95] M. Liu, S. Bell, M. Segarra, N.H. Steven Tay, G. Will, W. Saman, et al., A eutectic salt high temperature phase change material: thermal stability and corrosion of SS316 with respect to thermal cycling, Sol. Energy Mater. Sol. Cells 170 (2017) 1–7.
[96] C. Edeleanu, R. Littlewood, Thermodynamics of corrosion in fused chlorides, Electrochim. Acta 3 (1960) 195–207.
[97] Y. Li, C.G. Wang, G.Y. Liu, Q.Z. Zhu, Z.Z. Qiu, Thermal property characterization of a low supercooling degree binary mixed molten salt for thermal energy storage system, Int. J. Thermophys. 40 (2019) 41.
[98] M.M. Kenisarin, High-temperature phase change materials for thermal energy storage, Renew. Sust. Energ. Rev. 14 (2010) 955–970.
[99] P. Blanco-Rodríguez, J. Rodríguez-Aseguinolaza, E. Risueño, M. Tello, Thermophysical characterization of Mg–51%Zn eutectic metal alloy: a phase change material for thermal energy storage in direct steam generation applications, Energy 72 (2014) 414–420.
[100] T. Nomura, J. Yoolerd, N. Sheng, H. Sakai, Y. Hasegawa, M. Haga, et al., Microencapsulation of eutectic and hyper-eutectic Al-Si alloy as phase change materials for high-temperature thermal energy storage, Sol. Energy Mater. Sol. Cells 187 (2018) 255–262.
[101] Z. Wang, H. Wang, M. Yang, W. Sun, G. Yin, Q. Zhang, et al., Thermal reliability of Al-Si eutectic alloy for thermal energy storage, Mater. Res. Bull. 95 (2017) 300–306.
[102] T. Akiyama, Y. A, J. Yagi, Storage and release of heat in a single spherical capsule containing phase-change material with a high melting point, Heat Transf. Jap. Res. 21 (1992) 199–217.
[103] A.M. Gasanealiev, B.Y. G., Heat-accumulating properties of melts, Russ. Chem. Rev. 69 (2000) 179–186.
[104] J. Jiao, B. Grorud, C. Sindland, J. Safarian, K. Tang, K. Sellevoll, et al., The use of eutectic Fe-Si-B alloy as a phase change material in thermal energy storage systems, Materials (2019) 12.
[105] N. Xie, Z. Li, X. Gao, Y. Fang, Z. Zhang, Preparation and performance of modified expanded graphite/eutectic salt composite phase change cold storage material, Int. J. Refrig. 110 (2020) 178–186.
[106] K. Shahbaz, I.M. AlNashef, R.J.T. Lin, M.A. Hashim, F.S. Mjalli, M.M. Farid, A novel calcium chloride hexahydrate-based deep eutectic solvent as a phase change materials, Sol. Energy Mater. Sol. Cells 155 (2016) 147–154.
[107] G. Li, B. Zhang, X. Li, Y. Zhou, Q. Sun, Q. Yun, The preparation, characterization and modification of a new phase change material: CaCl2·6H2O–MgCl2·6H2O eutectic hydrate salt, Sol. Energy Mater. Sol. Cells 126 (2014) 51–55.
[108] C. Zhang, Z. Zhang, R. Ye, X. Gao, Z. Ling, Characterization of MgCl2·6H2O-based eutectic/expanded perlite composite phase change material with low thermal conductivity, Materials (2018) 11.
[109] O. Dong, D. Li, D. Zeng, A novel eutectic phase-change material: CaCl2·6H2O + NH4Cl + KCl, Calphad 63 (2018) 92–99.
[110] P. Ramanan, K. Kalidasa Murugavel, D. Hari Kishan, G. Suriyanarayanan, Temperature regulation of CIS photovoltaic module using eutectic salt hydrate as PCM, in: L.-J. Yang, A.N. Haq, L. Nagarajan (Eds.), Proceedings of ICDMC 2019, Springer Singapore, Singapore, 2020, pp. 637–652.
[111] S. Ushak, M. Vega, J.A. Lovera-Copa, S. Pablo, M. Lujan, M. Grageda, Thermodynamic modeling and experimental verification of new eutectic salt mixtures as thermal energy storage materials, Sol. Energy Mater. Sol. Cells 209 (2020), 110475.
[112] J.A. Lovera-Copa, S. Ushak, N. Reinaga, I. Villalobos, F.R. Martínez, Design of phase change materials based on salt hydrates for thermal energy storage in a range of 4–40 °C, J. Therm. Anal. Calorim. 139 (2020) 3701–3710.
[113] P. Hou, J. Mao, F. Chen, Y. Li, X. Dong, Preparation and thermal performance enhancement of low temperature eutectic composite phase change materials based on Na2SO4·10H2O, Materials. 11 (2018) 2230.
[114] Z. Rao, T. Xu, C. Liu, Z. Zheng, L. Liang, K. Hong, Experimental study on thermal properties and thermal performance of eutectic hydrated salts/expanded perlite form-stable phase change materials for passive solar energy utilization, Sol. Energy Mater. Sol. Cells 188 (2018) 6–17.
[115] N. Xie, J. Niu, Y. Zhong, X. Gao, Z. Zhang, Y. Fang, Development of polyurethane acrylate coated salt hydrate/diatomite form-stable phase change material with enhanced thermal stability for building energy storage, Constr. Build. Mater. 259 (2020), 119714.
[116] N. Xie, J. Luo, Z. Li, Z. Huang, X. Gao, Y. Fang, et al., Salt hydrate/expanded vermiculite composite as a form-stable phase change material for building energy storage, Sol. Energy Mater. Sol. Cells 189 (2019) 33–42.
[117] R. Pichandi, K. Murugavel Kulandaivelu, K. Alagar, H.K. Dhevaguru, S. Ganesamoorthy, Performance enhancement of photovoltaic module by integrating eutectic inorganic phase change material, Energy Sources A: Recovery Util. Environ. Eff. (2020) 1–18.
[118] W. Xin, J. Fang, W. Jiang, L. Ping, L. Na, F. Yanhan, et al., Preparation and modification of novel phase change material Na2SO4·10H2O-Na2HPO4·12H2O binary eutectic hydrate salt, Energy Sources A: Recovery Util. Environ. Eff. (2019) 1–12.
[119] Y. Liu, Y. Yang, Use of nano-α-Al2O3 to improve binary eutectic hydrated salt as phase change material, Sol. Energy Mater. Sol. Cells 160 (2017) 18–25.
[120] A. Karthick, K. Kalidasa Murugavel, A. Ghosh, K. Sudhakar, P. Ramanan, Investigation of a binary eutectic mixture of phase change material for building integrated photovoltaic (BIPV) system, Sol. Energy Mater. Sol. Cells 207 (2020), 110360.
[121] Z. Ling, J. Liu, Q. Wang, W. Lin, X. Fang, Z. Zhang, MgCl2·6H2O-Mg(NO3) 2·6H2O eutectic/SiO2 composite phase change material with improved thermal reliability and enhanced thermal conductivity, Sol. Energy Mater. Sol. Cells 172 (2017) 195–201.
[122] Y. Zhou, W. Sun, Z. Ling, X. Fang, Z. Zhang, Hydrophilic modification of expanded graphite to prepare a high-performance composite phase change block containing a hydrate salt, Ind. Eng. Chem. Res. 56 (2017) 14799–14806.
[123] A. Efimova, S. Pinnau, M. Mischke, C. Breitkopf, M. Ruck, P. Schmidt, Development of salt hydrate eutectics as latent heat storage for air conditioning and cooling, Thermochim. Acta 575 (2014) 276–278.
[124] T.V. Gubanova, I.M. Kondratyuk, I.K. Garkushin, Phase equilibria in the LiF-LiCl-LiVO3-Li2CrO 4 quaternary system, Russ. J. Inorg. Chem. 49 (2004) 1087–1090.
[125] T.V. Gubanova, I.K. Garkushin, LiF-LiCl-LiVO3-Li2SO4-Li 2MoO4 system, Russ. J. Inorg. Chem. 52 (2007) 1624–1628.
[126] T.V. Gubanova, I.M. Kondratyuk, I.K. Garkushin, The LiF-LiCl-Li2SO4-Li2MoO4 quaternary system, Russ. J. Inorg. Chem. 51 (2006) 474–477.
[127] W.M. Phillips, J.W. Stearns, Advanced latent heat of fusion thermal energy storage for solar power systems. Intersociety Energy Conversion Engineering Conference, 1985.
[128] K.E. Mayo, Heat Source Systems, 1970.
[129] T.V. Gubanova, E.I. Frolov, I.K. Garkushin, LiF-LiVO3-Li2SO4-Li2MoO 4 four-component system, Russ. J. Inorg. Chem. 52 (2007) 265–268.
[130] G.R. Heidenreich, M.B. Parekh, Thermal Energy Storage for Organic Rankine Cycle Solar Dynamic Space Power Systems, 1986, pp. 791–797.
[131] L.G. Marianowski, H.C. Maru, Latent heat thermal energy storage systems above 450 C, in: Intersociety Energy Conversion Engineering Conference, 1977, pp. 555–566.
[132] R.J. Petri, E.T. Ong, J. Martin, High Temperature Composite Thermal Energy Storage (TES) Systems for Industrial Applications, 1986.
[133] N.N. Verdiev, Y.A. Dibirov, P.A. Arbukhanova, S.I. Vainshtein, The three-component mutual system of Li, Ca//SO4, MoO 4, Mosc. Univ. Chem. Bull. 64 (2009) 111–116.
[134] Y. Abe, M. Kamimoto, Y. Takahashi, R. Sakamoto, K. Kanari, T. Ozawa, Peak load coverage by molten salts latent thermal storage, in: Proceedings of the Intersociety Energy Conversion Engineering Conference, 1984, pp. 1114–1119.
[135] A.F. Regin, S.C. Solanki, J.S. Saini, Heat transfer characteristics of thermal energy storage system using PCM capsules: a review, Renew. Sust. Energ. Rev. 12 (2008) 2438–2458.
[136] N.N. Verdiev, P.A. Arbukhanova, E.G. Iskenderov, NaF-NaBr-Na2MoO4 and KF-KCl-KBr ternary systems, Russ. J. Inorg. Chem. 54 (2009) 128–133.
[137] E.G. Iskenderov, N.N. Verdiev, S.I. Vainshtein, Phase equilibria in the NaCl-NaBr-Na2MoO4 system, Russ. J. Inorg. Chem. 52 (2007) 427–430.
[138] G. Wei, G. Wang, C. Xu, X. Ju, L. Xing, X. Du, et al., Selection principles and thermophysical properties of high temperature phase change materials for thermal energy storage: a review, Renew. Sust. Energ. Rev. 81 (2018) 1771–1786.
[139] M. Liu, W. Saman, F. Bruno, Review on storage materials and thermal performance enhancement techniques for high temperature phase change thermal storage systems, Renew. Sust. Energ. Rev. 16 (2012) 2118–2132.
[140] S. Li, S. Lin, Z. Ling, X. Fang, Z. Zhang, Growth of the phase change enthalpy induced by the crystal transformation of an inorganic–organic eutectic mixture of magnesium nitrate hexahydrate–glutaric acid, Ind. Eng. Chem. Res. 59 (2020) 6751–6760.
[141] P. Wang, X. Feng, Y. Zhu, J. Lian, H. Zhang, M. Fang, Preparation and thermal properties of colloidal mixtures of capric acid and Na2HPO4·12H2O as a phase change material for energy storage, Sol. Energy Mater. Sol. Cells 215 (2020), 110636.

[142] W. Sun, Q. Liu, J. Zhao, H. Muhammad Ali, Z. Said, C. Liu, Experimental study on sodium acetate trihydrate/glycerol deep eutectic solvent nanofluids for thermal energy storage, J. Mol. Liq. 372 (2023), 121164.
[143] H.M. Ali, An experimental study for thermal management using hybrid heat sinks based on organic phase change material, copper foam and heat pipe, J. Energy Storage 53 (2022), 105185.
[144] A. Hussain, H. Shahid, I. Ali, H.M. Ali, Passive thermal management system of lithium-ion batteries employing metal foam/ pcm composite for the development of electric vehicles, Energy Sources A: Recovery Util. Environ. Eff. 45 (2023) 505–522.
[145] Y.B. Tao, Y.-L. He, A review of phase change material and performance enhancement method for latent heat storage system, Renew. Sust. Energ. Rev. 93 (2018) 245–259.
[146] Z.A. Qureshi, H.M. Ali, S. Khushnood, Recent advances on thermal conductivity enhancement of phase change materials for energy storage system: a review, Int. J. Heat Mass Transf. 127 (2018) 838–856.
[147] Z. Huang, X. Gao, T. Xu, Y. Fang, Z. Zhang, Thermal property measurement and heat storage analysis of LiNO3/KCl – expanded graphite composite phase change material, Appl. Energy 115 (2014) 265–271.
[148] Z. Acem, Lopez Jm, E. Barrio, KNO3/NaNO3 – graphite materials for thermal energy storage at high temperature: part I. – elaboration methods and thermal properties, Appl. Therm. Eng. 30 (2010) 1580–1585.
[149] J. Lopez, Z. Acem, E. Barrio, KNO3/NaNO3 – graphite materials for thermal energy storage at high temperature: part II. – phase transition properties, Appl. Therm. Eng. 30 (2010) 1586–1593.
[150] N. Zhang, Y. Yuan, Y. Yuan, X. Cao, X. Yang, Effect of carbon nanotubes on the thermal behavior of palmitic–stearic acid eutectic mixtures as phase change materials for energy storage, Sol. Energy 110 (2014) 64–70.
[151] I.H. S., A.R. A., K. S., Bifunctional nanoencapsulated eutectic phase change material core with SiO 2 /SnO 2 nanosphere shell for thermal and electrical energy storage, Mater. Des. 154 (2018) 291–301.
[152] I. Baskar, M. Chellapandian, K. Jeyasubramanian, LA-PA eutectic/ nano- SiO2 composite phase change material for thermal energy storage application in buildings, Constr. Build. Mater. 338 (2022), 127663.
[153] F. Ye, Z. Ge, Y. Ding, J. Yang, Multi-walled carbon nanotubes added to Na2CO3/ MgO composites for thermal energy storage, Particuology. 15 (2014) 56–60.
[154] S.I. Hussain, A.A. Roseline, S. Kalaiselvam, Enhancement of thermal conductivity and thermal stability of capric-lauric acid eutectic phase change material using carbonaceous materials, Mater. Res. Express. 6 (2019) 115605.
[155] Z. Ling, S. Li, Z. Zhang, X. Fang, X. Gao, T. Xu, A shape-stabilized MgCl2·6H2O–Mg(NO3)2·6H2O/expanded graphite composite phase change material with high thermal conductivity and stability, J. Appl. Electrochem. 48 (2018) 1131–1138.
[156] Z. Chen, X. Zhang, J. Ji, Y. Lv, A review of the application of hydrated salt phase change materials in building temperature control, J. Energy Storage 56 (2022), 106157.
[157] Y. Li, C. Li, N. Lin, B. Xie, D. Zhang, J. Chen, Review on tailored phase change behavior of hydrated salt as phase change materials for energy storage, Materials Today Energy. 22 (2021), 100866.
[158] W.U.Z.J. Yu, W.A.N.G. Yongchao, X.I.E. Shuai, W.A.N.G. Jing, M.A. Chao, Preparation and performance analysis of sodium sulfate decahydrate/expanded vermiculite film phase change composite, Energy Storage Sci. Technol. 11 (2022) 3142–3150.
[159] T. Wu, N. Xie, J. Niu, J. Luo, X. Gao, Y. Fang, et al., Preparation of a low-temperature nanofluid phase change material: MgCl2–H2O eutectic salt solution system with multi-walled carbon nanotubes (MWCNTs), Int. J. Refrig. 113 (2020) 136–144.
[160] Y. Song, B. Eck, J. Levi, D. Wilson, Improved Accuracy and Precision of Calcium Mass Score Using Deconvolution and Partial Volume Correction, 2020.
[161] Y. Zhao, H.B. Liu, C.Y. Zhao, Experimental study on the cycling stability and corrosive property of Al-Si alloys as phase change materials in high-temperature heat storage, Sol. Energy Mater. Sol. Cells 203 (2019), 110165.
[162] G. Ferrer, A. Solé, C. Barreneche, I. Martorell, L.F. Cabeza, Corrosion of metal containers for use in PCM energy storage, Renew. Energy 76 (2015) 465–469.
[163] M.C. Browne, E. Boyd, S.J. McCormack, Investigation of the corrosive properties of phase change materials in contact with metals and plastic, Renew. Energy 108 (2017) 555–568.
[164] M. Liu, X. Zhang, J. Ji, H. Yan, Review of research progress on corrosion and anti-corrosion of phase change materials in thermal energy storage systems, J. Energy Storage 63 (2023), 107005.
[165] A. Karaipekli, A. Sar, Capric-myristic acid/vermiculite composite as form-stable phase change material for thermal energy storage, Sol. Energy 323-32 (2009).
[166] Eanest B. Jebasingh, Effects of various types of graphite on the thermal conductivity and energy storage properties of ternary eutectic fatty acid-based composite as phase change material, Renewables Wind Water Solar 3 (2016) 8.
[167] X. Zhang, Z. Huang, Z. Yin, W. Zhang, Y. Huang, Y. Liu, et al., Form stable composite phase change materials from palmitic-lauric acid eutectic mixture and carbonized abandoned rice: preparation, characterization, and thermal conductivity enhancement, Energy Build. S0378778817318479 (2017).
[168] S. Dhivya, S.I. Hussain, S. Kalaiselvam, Novel metal coated nanocapsules of ethyl esters fatty acid eutectic mixture as phase change material with enhanced thermal conductivity for energy storage applications, Thermochim. Acta 687 (2020), 178581.
[169] G. Ma, Z. Wang, S. Xie, J. Sun, Y. Jia, Y. Jing, et al., Preparation and properties of stearic acid-acetanilide eutectic mixture/expanded graphite composite phase-change material for thermal energy storage, Energy Technol. 6 (2018) 153–160.
[170] Y. Seki, İ. Şeyma, M.A. Ezan, A. Turgut, A. Erek, Graphite nanoplates loading into eutectic mixture of Adipic acid and Sebacic acid as phase change material, Sol. Energy Mater. Sol. Cells 140 (2015) 457–463.
[171] T. Wang, Y. Liu, R. Meng, M. Zhang, Thermal performance of galactitol/mannitol eutectic mixture/expanded graphite composite as phase change material for thermal energy harvesting, J. Energy Storage (2021) 34.
[172] L. Abdolmaleki, S.M. Sadrameli, A. Pirvaram, Application of environmental friendly and eutectic phase change materials for the efficiency enhancement of household freezers, Renew. Energy 145 (2020) 233–241.
[173] B. Gin, M.M. Farid, P.K. Bansal, Effect of door opening and defrost cycle on a freezer with phase change panels, Energy Convers. Manag. 51 (2010) 2698–2706.
[174] J. Cofré-Toledo, D.A. Vasco, C.A. Isaza-Roldán, J.A. Tangarife, Evaluation of an integrated household refrigerator evaporator with two eutectic phase-change materials, Int. J. Refrig. 93 (2018) 29–37.
[175] Y. Zhao, X. Zhang, X. Xu, S. Zhang, Development of composite phase change cold storage material and its application in vaccine cold storage equipment, J. Energy Storage 30 (2020), 101455.
[176] L. Liu, X. Zhang, X. Xu, X. Lin, Y. Zhao, L. Zou, et al., Development of low-temperature eutectic phase change material with expanded graphite for vaccine cold chain logistics, Renew. Energy 179 (2021) 2348–2358.
[177] S.C. Turner, E.A. Arens, R.M. Aynsley, G.S. Brager, J.J. Deringer, J.M. Ferguson, et al., ASHRAE STANDARD Thermal Environmental Conditions for Human Occupancy, 2010.
[178] C. Barreneche, M.E. Navarro, S. Serrano, L.F. Cabeza, A. Fernández, New database on phase change materials for thermal energy storage in buildings to help PCM selection, Energy Procedia 57 (2014) 2408–2415.
[179] Chinnasamy Veerakumar, Appukuttan Sreekumar, Preparation, thermophysical studies, and corrosion analysis of a stable capric acid/cetyl alcohol binary eutectic phase change material for cold thermal energy storage, Energy Technol. Gener. Convers. Storage Distrib. 6 (2018) 397–405.
[180] R. Ye, H. Jiang, J. Wang, X. Yang, X. Shu, Fabrication and characteristics of eutectic hydrated salts/fumed silica composite as form-stable phase change materials for thermal energy storage, Sol. Energy Mater. Sol. Cells 238 (2022), 111584.
[181] X. Zhang, L. Guo, J. Ren, X. Kong, Characterization of expanded graphite-based erythritol/urea eutectic phase change material and corresponding mathematical solar heating application analysis, Appl. Therm. Eng. 215 (2022), 118938.
[182] K. Ma, X. Zhang, J. Ji, L. Han, X. Ding, W. Xie, Application and research progress of phase change materials in biomedical field, Biomater. Sci. 9 (2021) 5762–5780.
[183] C. Zhu, D. Huo, Q. Chen, J. Xue, S. Shen, Y. Xia, A eutectic mixture of natural fatty acids can serve as the gating material for near-infrared-triggered drug release, Adv. Mater. 29 (2017).
[184] B. Hemmatian, N. Heidarzadeh, G. Chizari Fard, L. Maleknia, Fabrication of phase-change core/shell nanofibers based on a eutectic fatty acid mixture to control body temperature fluctuations, Mater. Chem. Phys. 245 (2020), 122738.
[185] J. Xu, Y. Zhou, Z. Xiong, H. OuYang, F. Huang, Y. Yang, et al., Hyperthermia-triggered UK release nanovectors for deep venous thrombosis therapy, J. Mater. Chem. B (2019) 8.
[186] A. Karthick, K.K. Murugavel, K. Sudalaiyandi, A.M. Manokar, Building integrated photovoltaic modules and the integration of phase change materials for equatorial applications, Build. Serv. Eng. Res. Technol. 41 (2020) 634–652.
[187] K. Velmurugan, V. Karthikeyan, T.B. Korukonda, P. Poongavanam, D. Sandeep, Experimental studies on photovoltaic module temperature reduction using eutectic cold phase change material, Sol. Energy 209 (2020) 302–315.
[188] M. Gurturk, A. Koca, H.F. Otztop, Y. Varol, M. Sekerci, Energy and exergy analysis of a heat storage tank with a novel eutectic phase change material layer of a solar heater system, Int. J. Green Energy 14 (2017) 1073–1080.
[189] A. Sar, Eutectic mixtures of some fatty acids for latent heat storage: thermal properties and thermal reliability with respect to thermal cycling, Energy Convers. Manag. 47 (2006) 1207–1221.
[190] S. Keleg, K. Kaygusuz, A. San, Laurie and myristic acids eutectic mixture as phase change material for low-temperature heating applications, Int. J. Energy Res. 29 (2005) 857–870.
[191] C. Alkan, E. Günther, S. Hiebler, Ö.F. Ensari, D. Kahraman, Polyurethanes as solid–solid phase change materials for thermal energy storage, Sol. Energy 86 (2012) 1761–1769.
[192] A.-M.A. El-Bassuoni, A.M. Tayeb, N.H. Helwa, A.M. Fathy, Modification of urea–sodium acetate trihydrate mixture for solar energy storage, Renew. Energy 28 (2003) 1629–1643.
[193] Z. Ling, S. Li, C. Cai, S. Lin, X. Fang, Z. Zhang, Battery thermal management based on multiscale encapsulated inorganic phase change material of high stability, Appl. Therm. Eng. 193 (2021), 117002.
[194] A. Rk, A. Ya, A. Yy, B. St, A. Fe, A. Ah, et al., Thermophysical properties of trimethylolethane (TME) hydrate as phase change material for cooling lithium-ion battery in electric vehicle, J. Power Sources 427 (2019) 70–76.
[196] K. Purohit, V.V.S. Murty, R.C. Dixit, A. Sharma, Development of an acetanilide/ benzoic acid eutectic phase change material based thermal energy storage unit for a passive water heating system, Bull. Mater. Sci. 42 (2019) 1–9.
[197] X. Tang, B. Zhu, M. Xu, W. Zhang, Z. Yang, Y. Zhang, et al., Shape-stabilized phase change materials based on fatty acid eutectics/expanded graphite composites for thermal storage, Energy Build. 109 (2015) 353–360.
[198] H.M. Ali, Phase change materials based thermal energy storage for solar energy systems, J. Build. Eng. 56 (2022), 104731.

[199] E. Risueño, S. Doppiu, J. Rodríguez-Aseguinolaza, P. Blanco, A. Gil, M. Tello, et al., Experimental investigation of Mg-Zn-Al metal alloys for latent heat storage application, J. Alloys Compd. 685 (2016) 724–732.
[200] Peng, X.X. Yang, Ding, X.L. Wei, et al., Design of new molten salt thermal energy storage material for solar thermal power plant, Appl. Energ. 2013 (112) (2013) 682–689.
[201] A. Fernández, A. Rey, I. Lasanta, S. Mato, M.P. Brady, F. Pérez, Corrosion of alumina-forming austenitic steel in molten nitrate salts by gravimetric analysis and impedance spectroscopy, Mater. Corros. 65 (2014) 267–275.
[202] P. Qiang, D. Jing, X. Wei, J. Yang, X. Yang, The preparation and properties of multi-component molten salts, Appl. Energy 87 (2010) 2812–2817.
[203] Q. Peng, X. Wei, J. Ding, J. Yang, X. Yang, High-temperature thermal stability of molten salt materials, Int. J. Energy Res. 32 (2008) 1164–1174.
[204] J.C. Gomez, N. Calvet, A.K. Starace, G.C. Glatzmaier, Ca(NO)—NaNO—KNOMolten salt mixtures for direct thermal energy storage systems in parabolic trough plants, J. Sol. Energy Eng. 135 (2013) 021016.1–021016..8.
[205] S.H. Goods, R.W. Bradshaw, Corrosion of stainless steels and carbon steel by molten mixtures of commercial nitrate salts, J. Mater. Eng. Perform. 13 (2004) 78–87.
[206] R.I. Dunn, P.J. Hearps, M.N. Wright, Molten-salt power towers: newly commercial concentrating solar storage, Proc. IEEE 100 (2012) 504–515.
[207] Y. Jiang, Y. Sun, R.D. Jacob, F. Bruno, S. Li, Novel Na2SO4-NaCl-ceramic composites as high temperature phase change materials for solar thermal power plants (part I), Sol. Energy Mater. Sol. Cells 178 (2018) 74–83.
[208] H. Peyrovedin, R. Haghbakhsh, A.R.C. Duarte, A. Shariati, Deep eutectic solvents as phase change materials in solar thermal power plants: energy and exergy analyses, Molecules 27 (2022) 1427.
[209] B. Hemmatian, N. Heidarzadeh, G.C. Fard, L. Maleknia, Fabrication of phase-change core/shell nanofibers based on a eutectic fatty acid mixture to control body temperature fluctuations, Mater. Chem. Phys. 245 (2020) 122738.
[210] R. Koyama, Y. Arai, Y. Yamauchi, S. Takeya, F. Endo, A. Hotta, et al., Thermophysical properties of trimethylolethane (TME) hydrate as phase change material for cooling lithium-ion battery in electric vehicle, J. Power Sources 427 (2019) 70–76.
[211] M. Sun, T. Liu, M. Li, J. Tan, P. Tian, H. Wang, et al., A deep supercooling eutectic phase change material for low-temperature battery thermal management, J. Energy Storage 50 (2022), 104240.